

Comparison Between RS67 Vs RS59

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April 24, 2026

Slide 02: Input Files (Roadset & Target Order)

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

Remark: Currently we don't process RS5a, RS5b files.

Slide 03: Physics Constants & Normalizations

Parameter	Value
PoT LH ₂	2.7639×10^{17}
PoT LD ₂	1.3197×10^{17}
PoT Flask	5.5905×10^{16}
LH ₂ Target Density	3.5966 mol/cm ²
LD ₂ Target Density	8.0431 mol/cm ²
LH ₂ / LD ₂ Target Length	50.8 cm
Avogadro Constant	6.022×10^{23}
xF Bin Width	0.05
Nuclear Int. Length (LH ₂ /LD ₂)	52.0 / 54.7 g/cm ²

Slide 04: Event Selection Summary

Dimuon Level $|dx| < 0.25$, $\text{mass} \in [4.2, 8.8]$, $x_F \in [-0.1, 0.95]$, $x_T \in [0.05, 0.58]$, $\chi^2 < 18$.

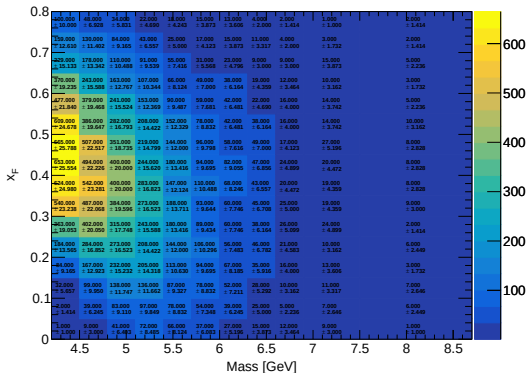
Track Level $\chi^2_{\text{target}} < 15$, $p_{z_{st1}} \in [9, 75]$, Hits > 13 , Station 1/3 y-ratio < 1 .

Occupancy $D1, D2, D3 < 400$ hits each; Total $D1 + D2 + D3 < 1000$.

Physics $Z_{\text{vertex}} \in [-320, -5]$, Track Separation < 270 , $|\cos \theta| < 0.5$.

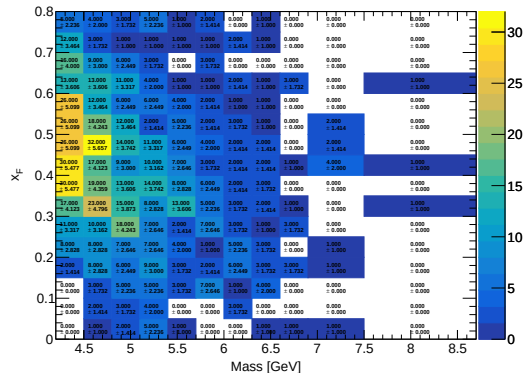
RS67

Total Yield (result) (LH2)



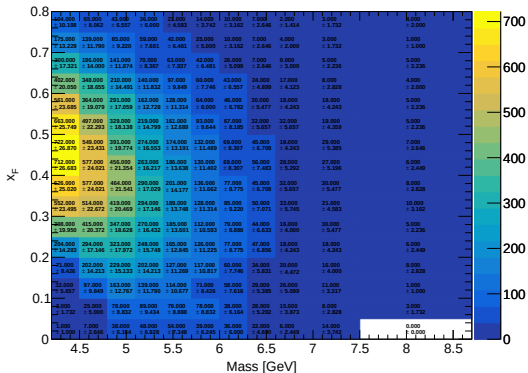
RS59

Total Yield (result) (LH2)



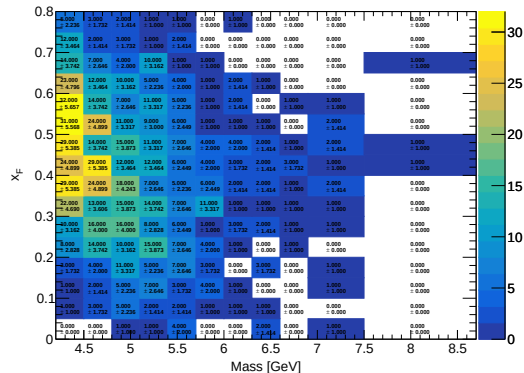
RS67

Total Yield (result) (LD2)



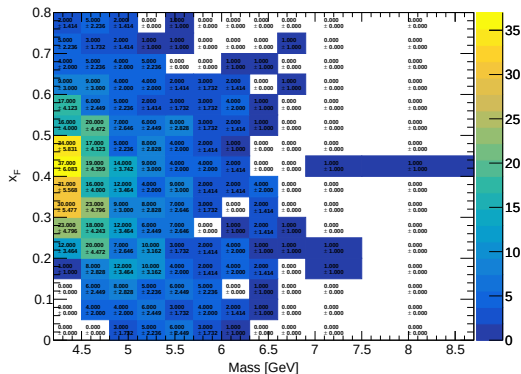
RS59

Total Yield (result) (LD2)



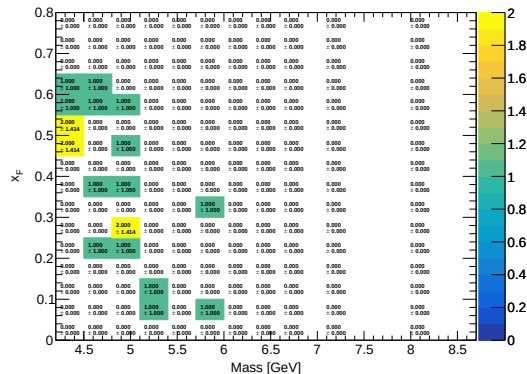
RS67

Total Yield (result) (Flask)



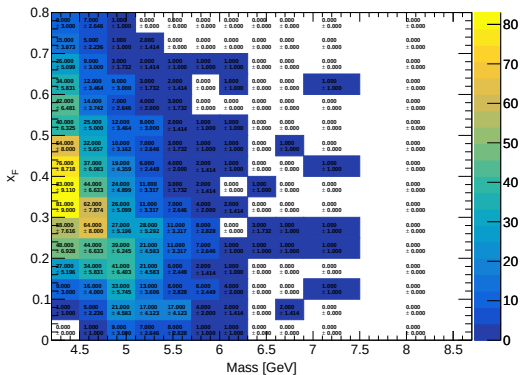
RS59

Total Yield (result) (Flask)



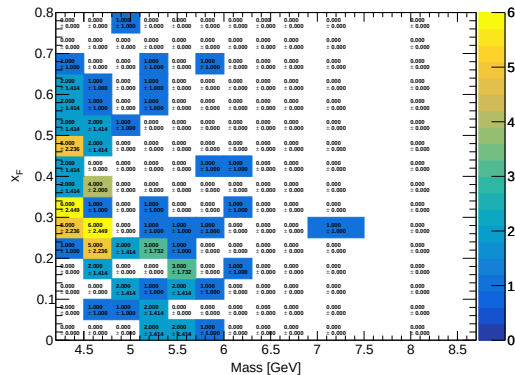
RS67

Mix Yield (result_mix) (LH2)



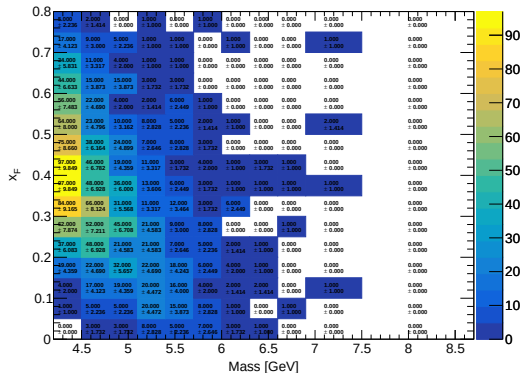
RS59

Mix Yield (result_mix) (LH2)



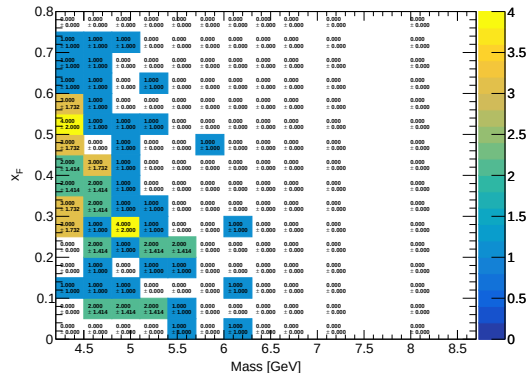
RS67

Mix Yield (result_mix) (LD2)



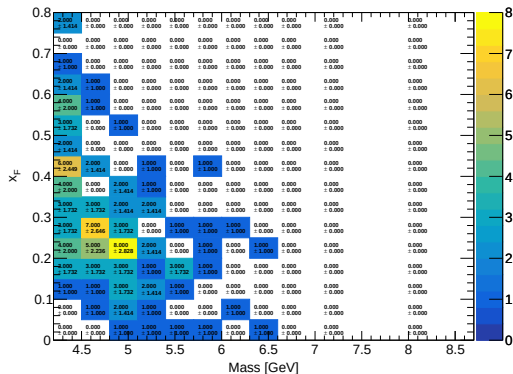
RS59

Mix Yield (result_mix) (LD2)



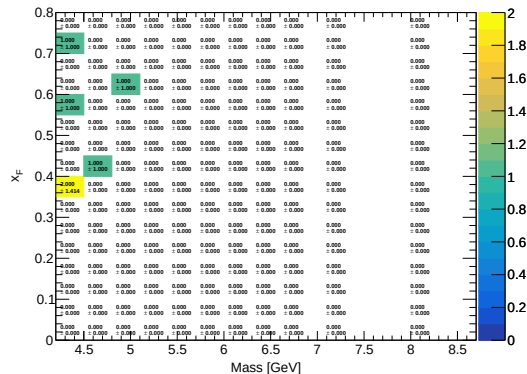
RS67

Mix Yield (result_mix) (Flask)



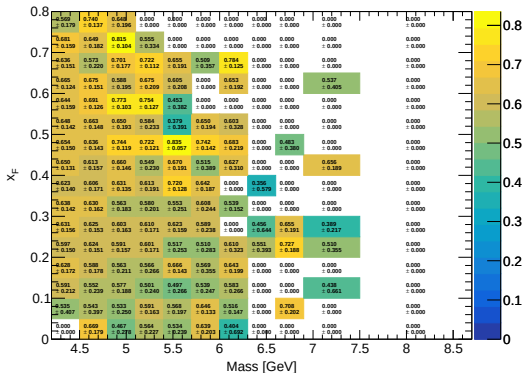
RS59

Mix Yield (result_mix) (Flask)



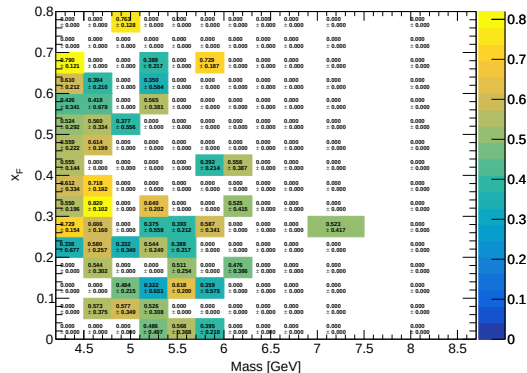
RS67

Avg Reco Eff (Mix) (LH2)



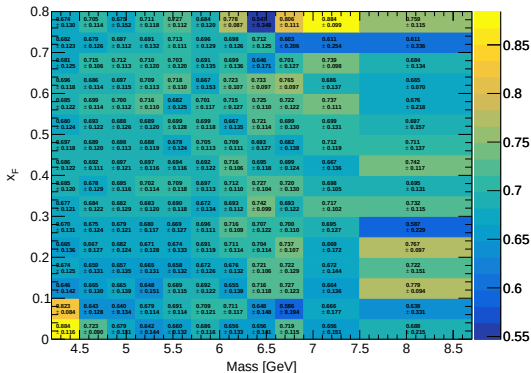
RS59

Avg Reco Eff (Mix) (LH2)



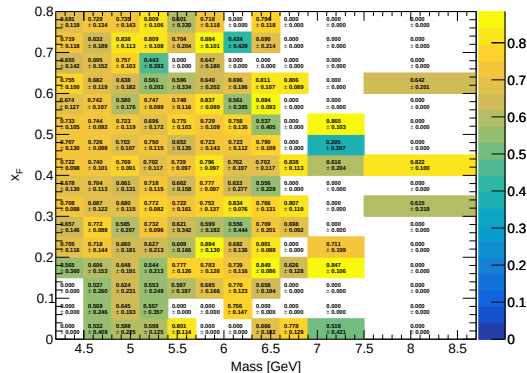
RS67

Avg Reco Eff (Total) (LH2)



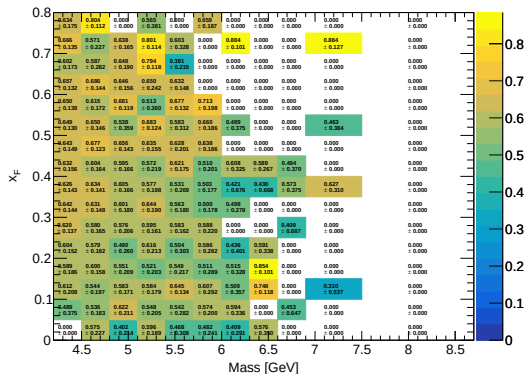
RS59

Avg Reco Eff (Total) (LH2)



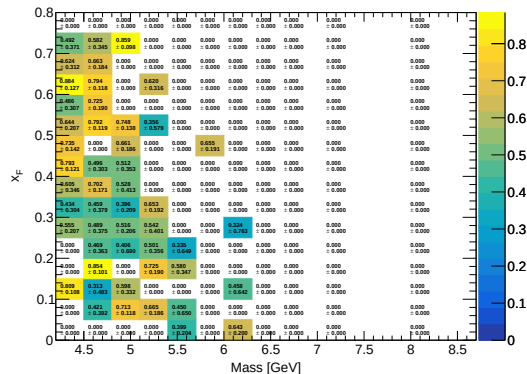
RS67

Avg Reco Eff (Mix) (LD2)



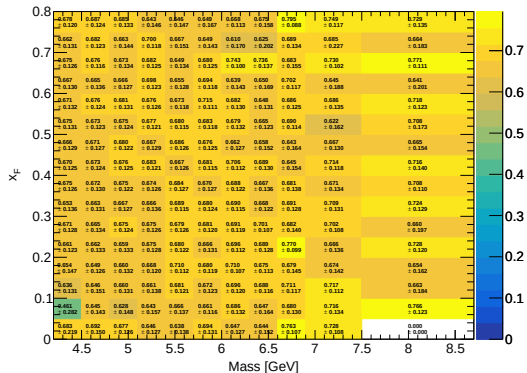
RS59

Avg Reco Eff (Mix) (LD2)



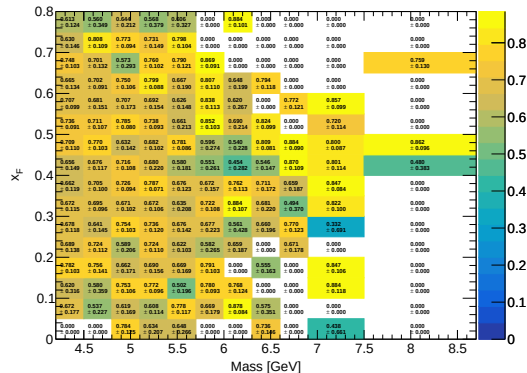
RS67

Avg Reco Eff (Total) (LD2)

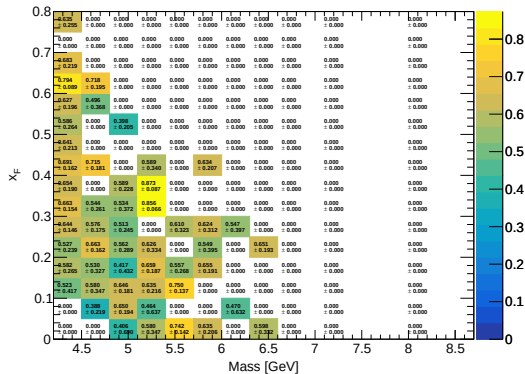


RS59

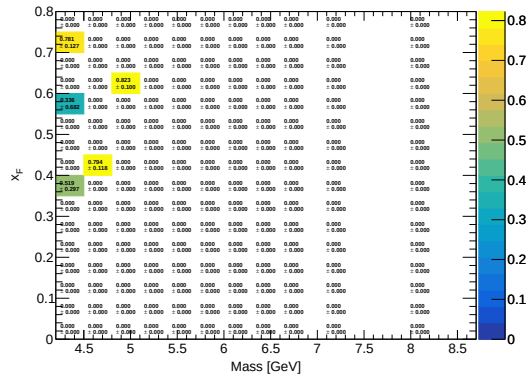
Avg Reco Eff (Total) (LD2)



RS67
Avg Reco Eff (Mix) (Flask)

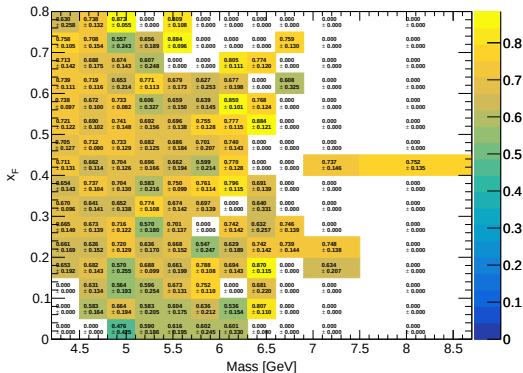


RS59
Avg Reco Eff (Mix) (Flask)



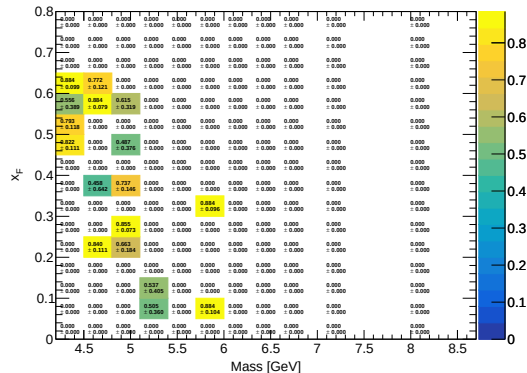
RS67

Avg Reco Eff (Total) (Flask)

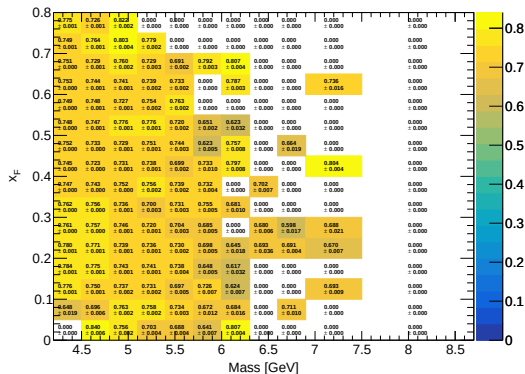


RS59

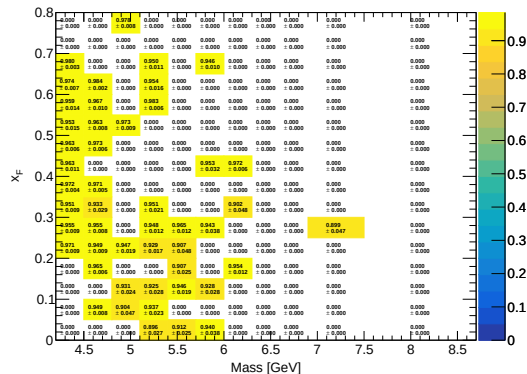
Avg Reco Eff (Total) (Flask)



RS67
Avg Hodo Eff (Mix) (LH2)

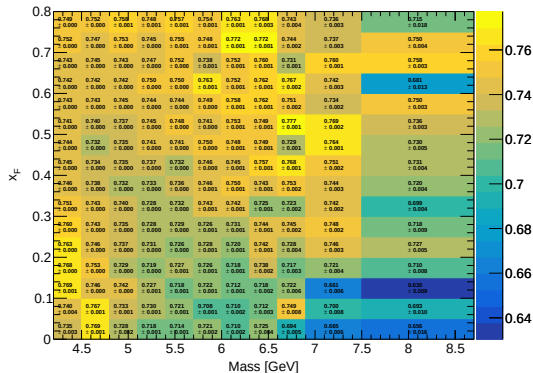


RS59
Avg Hodo Eff (Mix) (LH2)



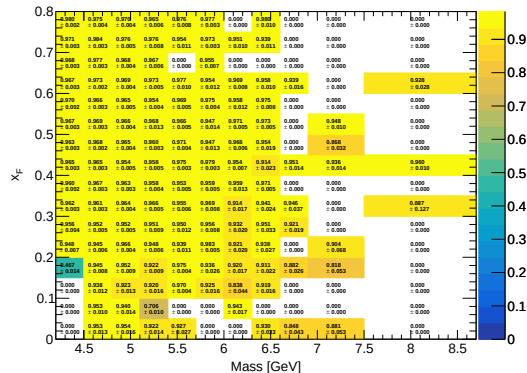
RS67

Avg Hodo Eff (Total) (LH2)

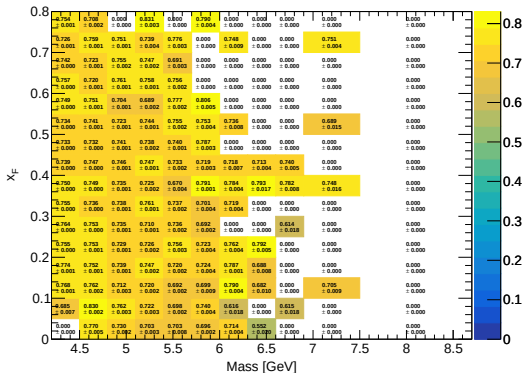


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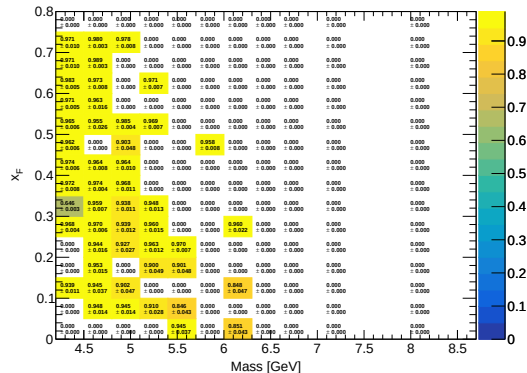
Avg Hodo Eff (Total) (LH2)



RS67
Avg Hodo Eff (Mix) (LD2)

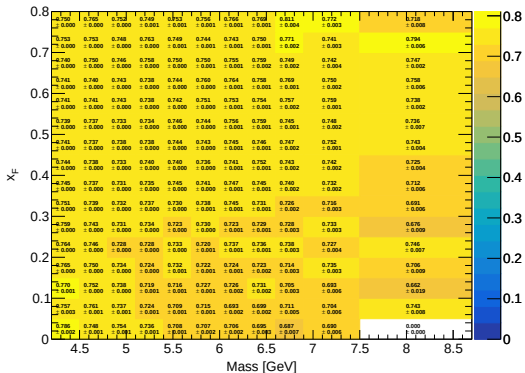


RS59
Avg Hodo Eff (Mix) (LD2)



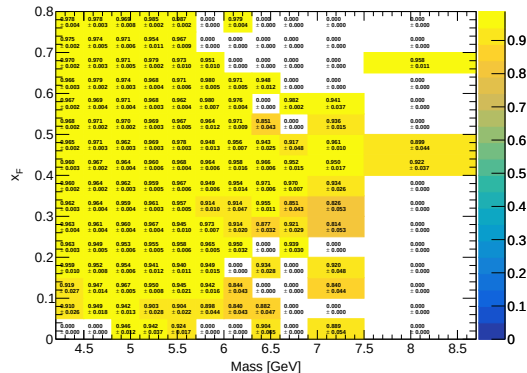
RS67

Avg Hodo Eff (Total) (LD2)

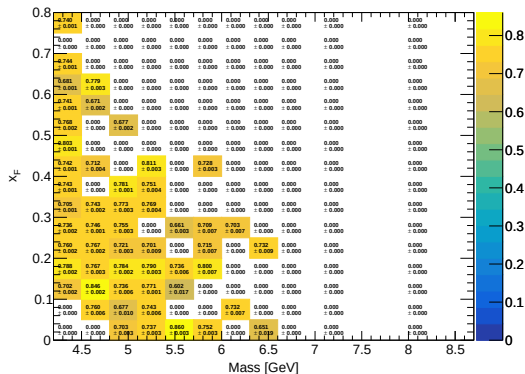


RS59

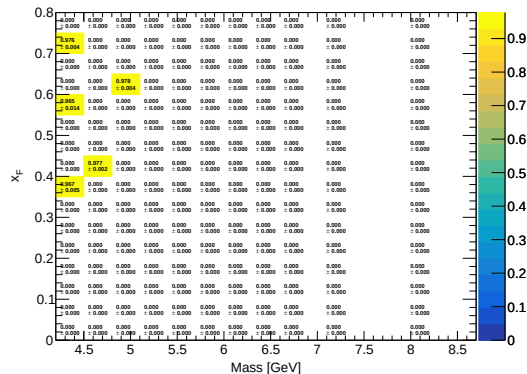
Avg Hodo Eff (Total) (LD2)



RS67
Avg Hodo Eff (Mix) (Flask)



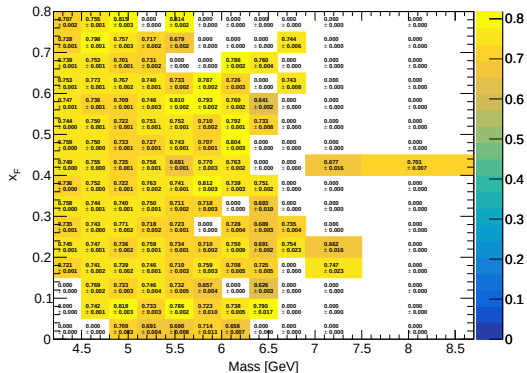
RS59
Avg Hodo Eff (Mix) (Flask)



E_total_hodo_Flask

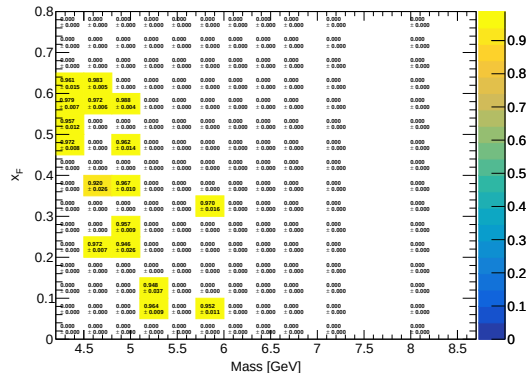
RS67

Avg Hodo Eff (Total) (Flask)



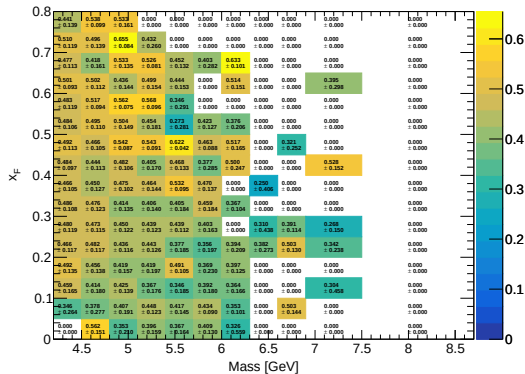
RS59

Avg Hodo Eff (Total) (Flask)



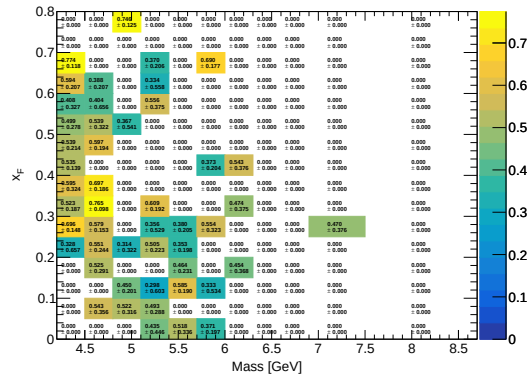
RS67

Avg Final Eff (Mix) (LH2)



RS59

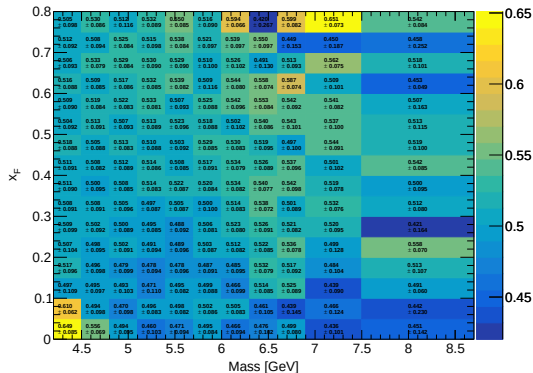
Avg Final Eff (Mix) (LH2)



E_total_final_LH2

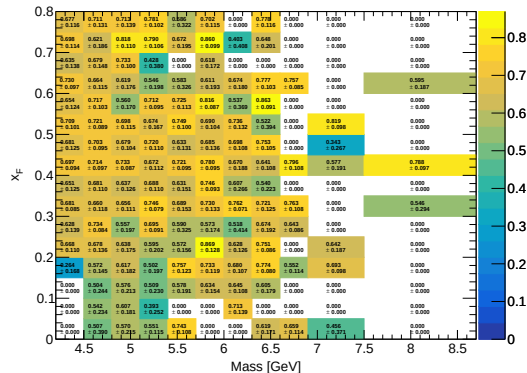
RS67

Avg Final Eff (Total) (LH2)



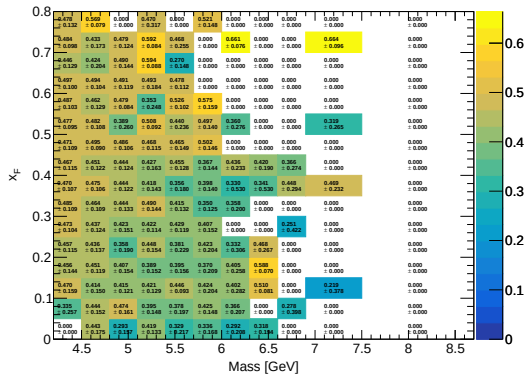
RS59

Avg Final Eff (Total) (LH2)



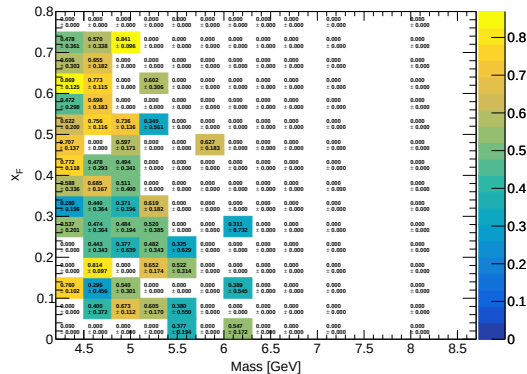
RS67

Avg Final Eff (Mix) (LD2)



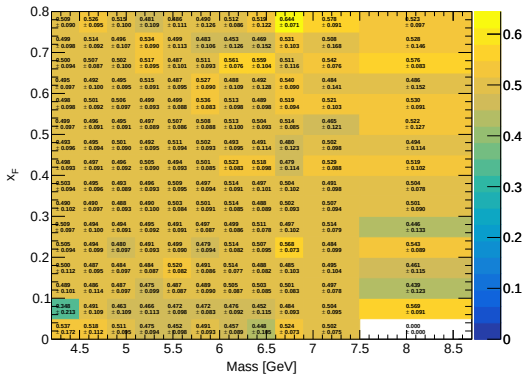
RS59

Avg Final Eff (Mix) (LD2)



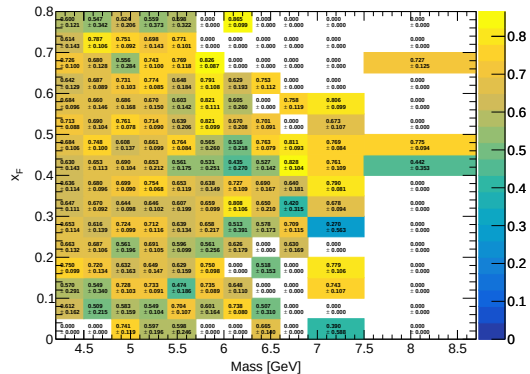
RS67

Avg Final Eff (Total) (LD2)

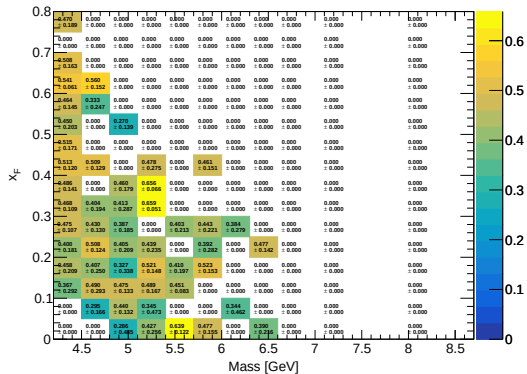


RS59

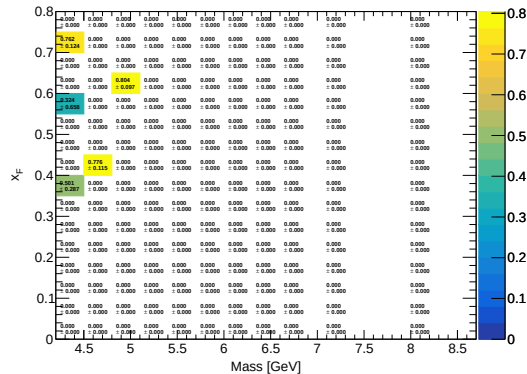
Avg Final Eff (Total) (LD2)



RS67
Avg Final Eff (Mix) (Flask)



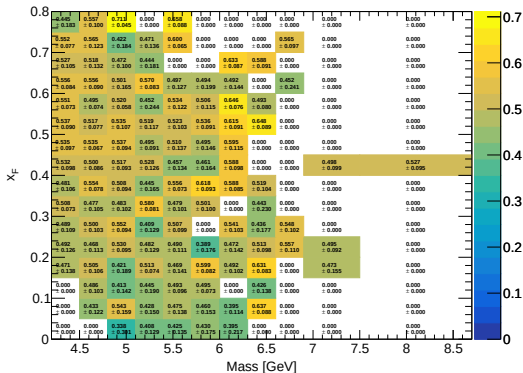
RS59
Avg Final Eff (Mix) (Flask)



E_total_final_Flask

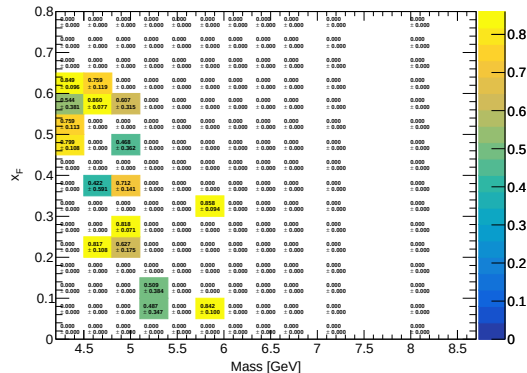
RS67

Avg Final Eff (Total) (Flask)



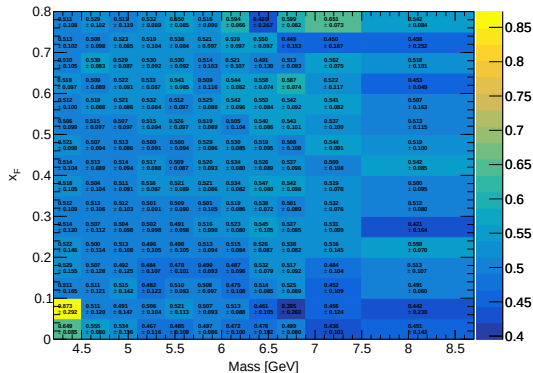
RS59

Avg Final Eff (Total) (Flask)



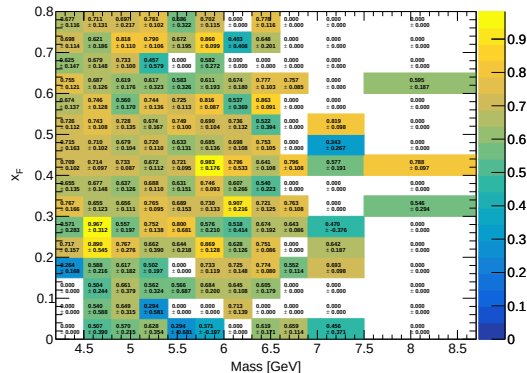
RS67

Avg Signal Efficiency (LH2)

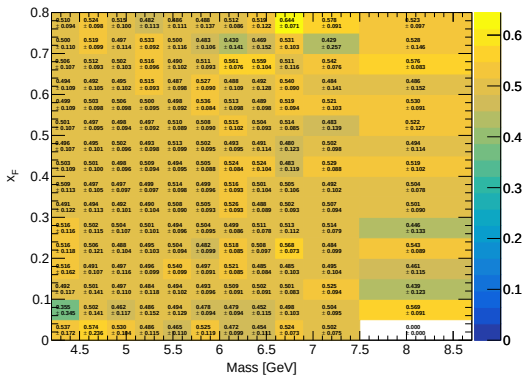


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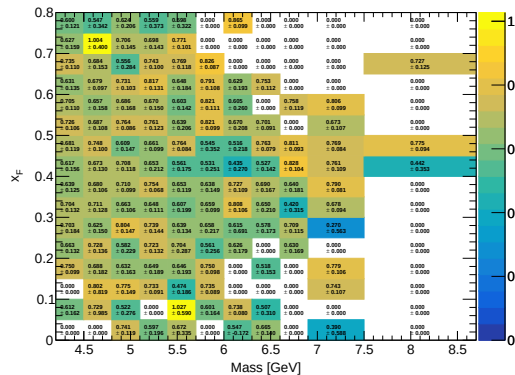
Avg Signal Efficiency (LH2)



RS67
Avg Signal Efficiency (LD2)

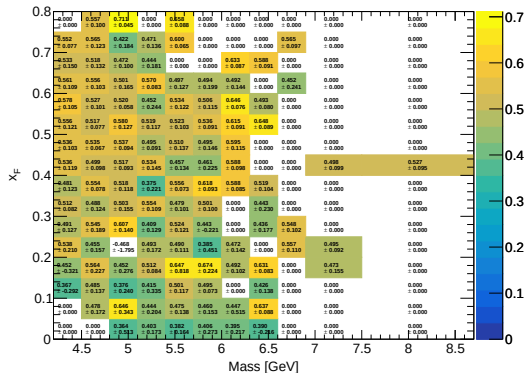


RS59
Avg Signal Efficiency (LD2)



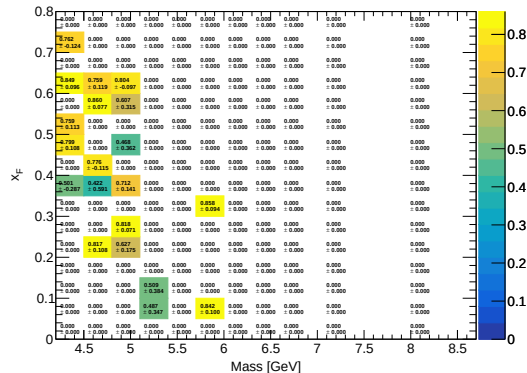
RS67

Avg Signal Efficiency (Flask)



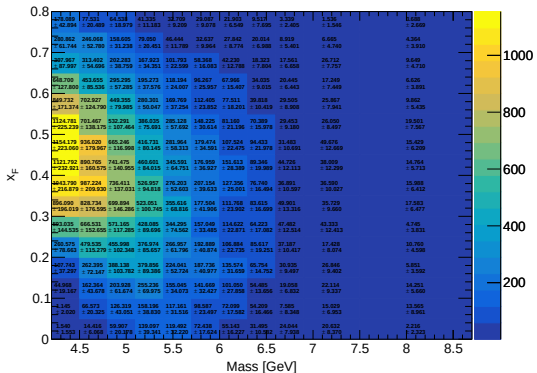
RS59

Avg Signal Efficiency (Flask)



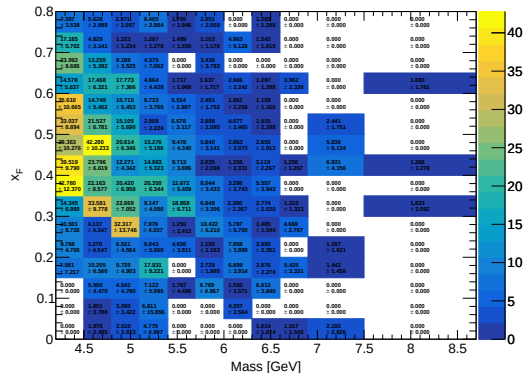
RS67

Corrected Yield (Total Error) (LH2)



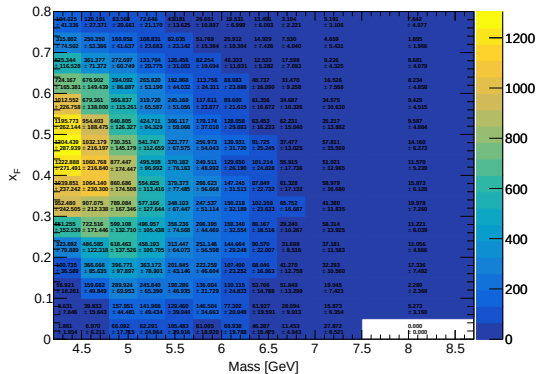
RS59

Corrected Yield (Total Error) (LH2)



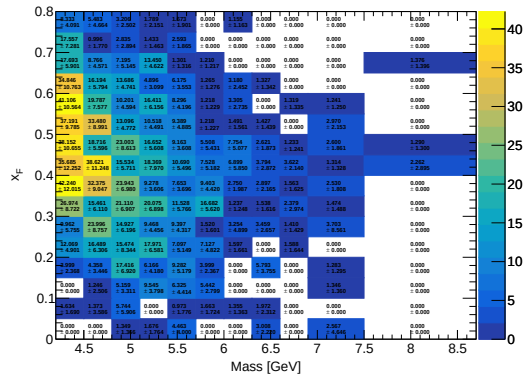
RS67

Corrected Yield (Total Error) (LD2)



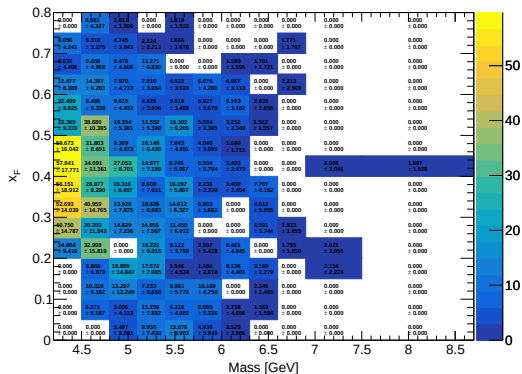
RS59

Corrected Yield (Total Error) (LD2)



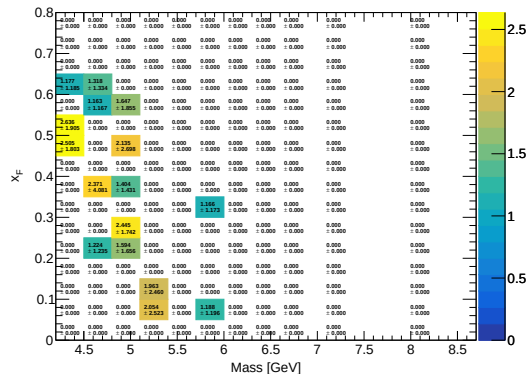
RS67

Corrected Yield (Total Error) (Flask)



RS59

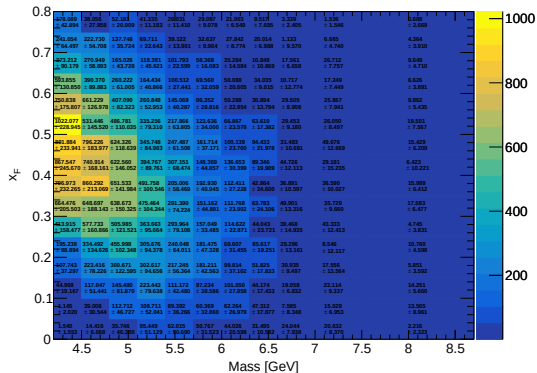
Corrected Yield (Total Error) (Flask)



Y_corrected_Subtracted_LH2

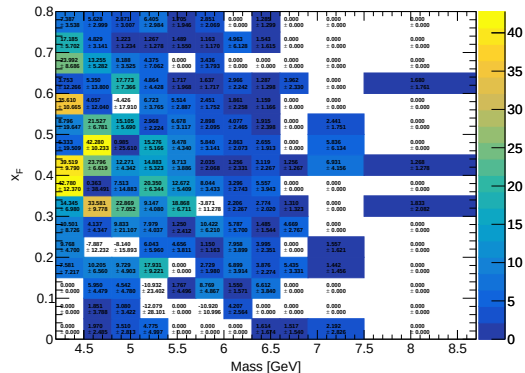
RS67

Corrected Yield (LH2 - Flask)



RS59

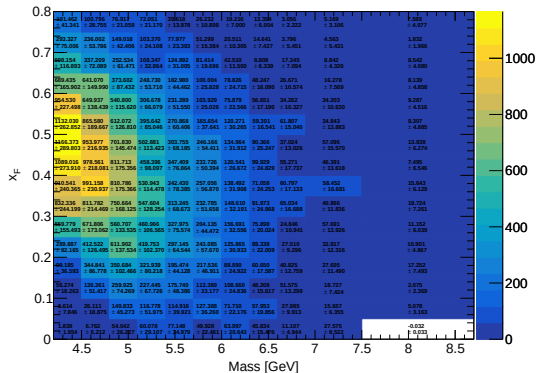
Corrected Yield (LH2 - Flask)



Y_corrected_Subtracted_LD2

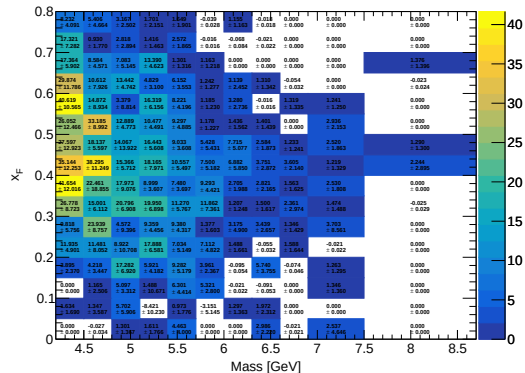
RS67

Corrected Yield (LD2 - LH2 - Flask)



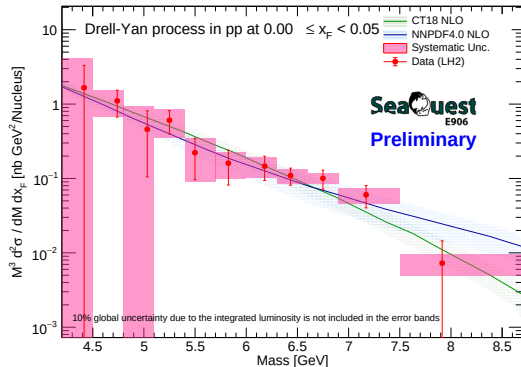
RS59

Corrected Yield (LD2 - LH2 - Flask)

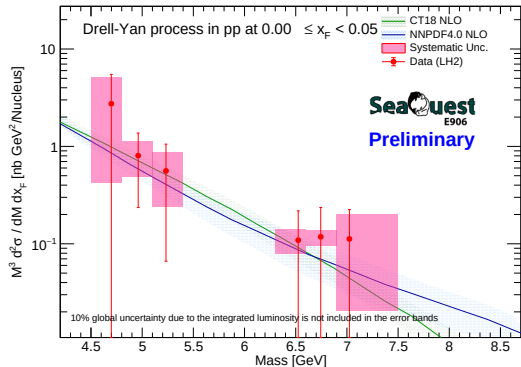


CrossSection LH2 $0.00 \leq x_F < 0.05$ (Centroid)

RS67

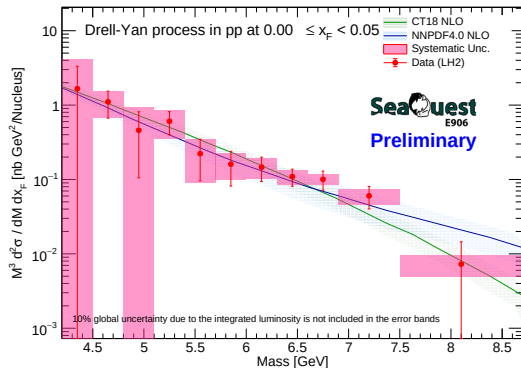


RS59

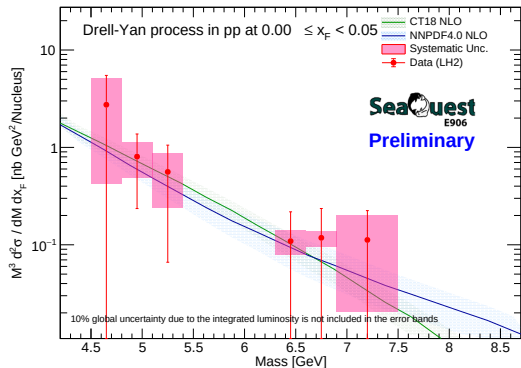


CrossSection LH2 $0.00 \leq x_F < 0.05$ (GeoCenter)

RS67

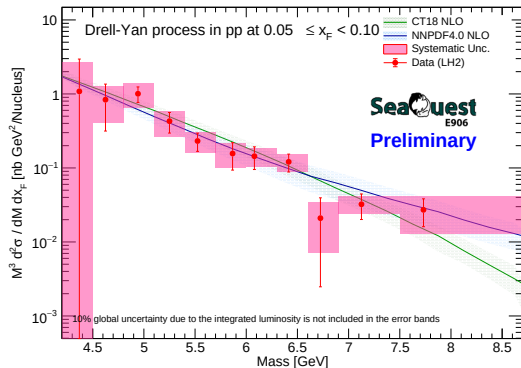


RS59

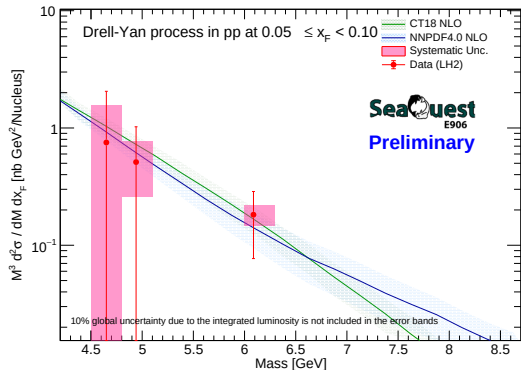


CrossSection LH2 $0.05 \leq x_F < 0.10$ (Centroid)

RS67

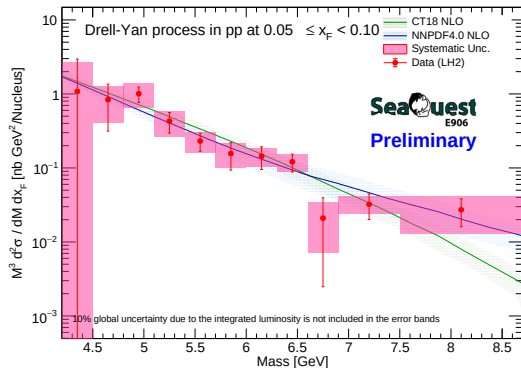


RS59

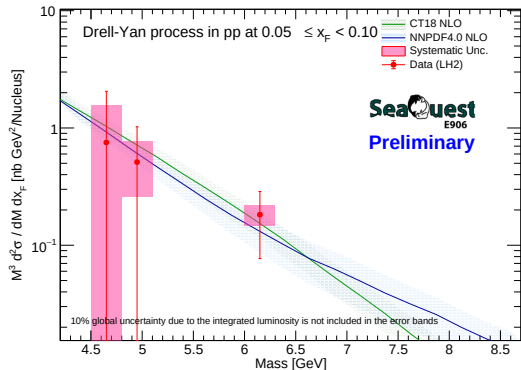


CrossSection LH2 $0.05 \leq x_F < 0.10$ (GeoCenter)

RS67

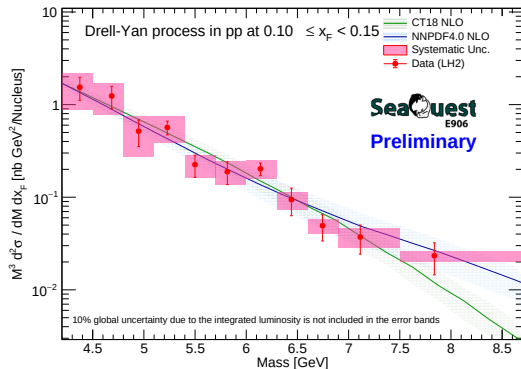


RS59

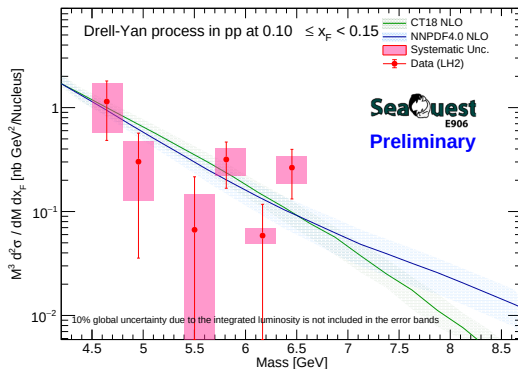


CrossSection LH2 $0.10 \leq x_F < 0.15$ (Centroid)

RS67

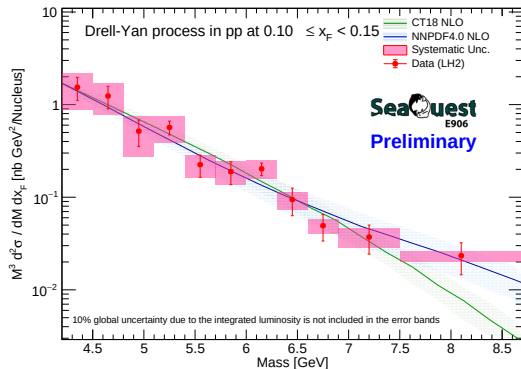


RS59

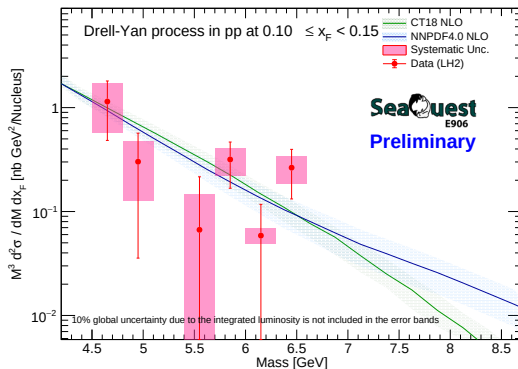


CrossSection LH2 $0.10 \leq x_F < 0.15$ (GeoCenter)

RS67

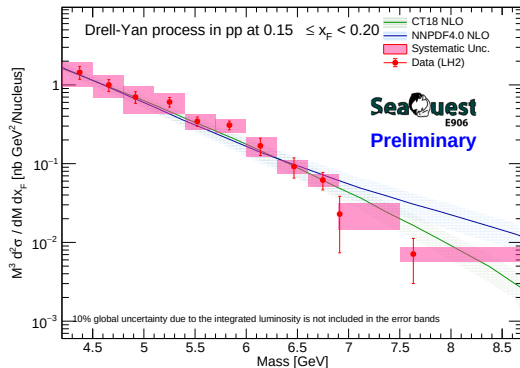


RS59

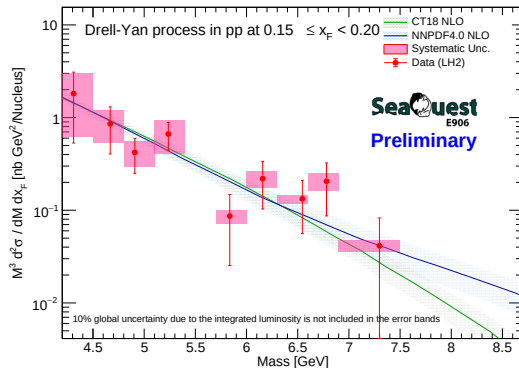


CrossSection LH2 $0.15 \leq x_F < 0.20$ (Centroid)

RS67

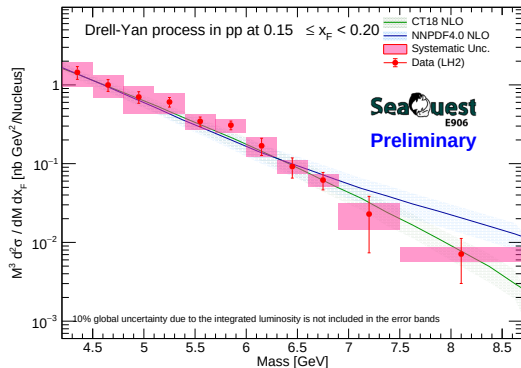


RS59

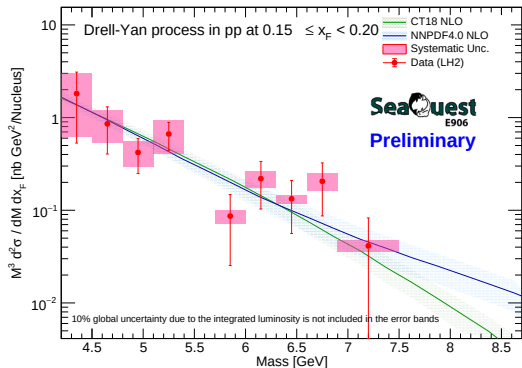


CrossSection LH2 $0.15 \leq x_F < 0.20$ (GeoCenter)

RS67

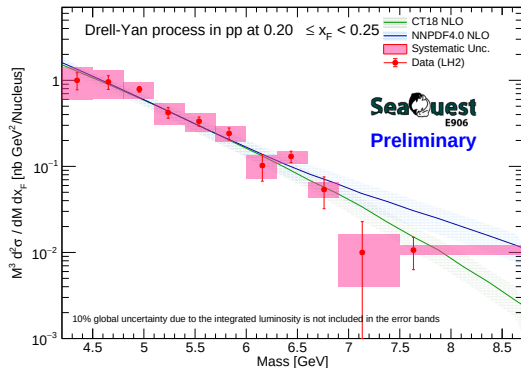


RS59

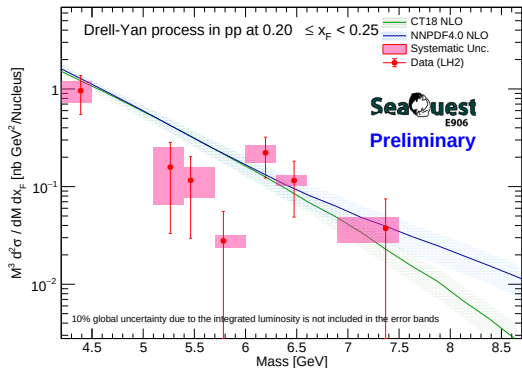


CrossSection LH2 $0.20 \leq x_F < 0.25$ (Centroid)

RS67

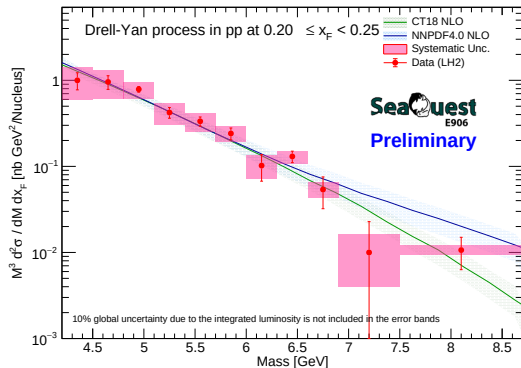


RS59

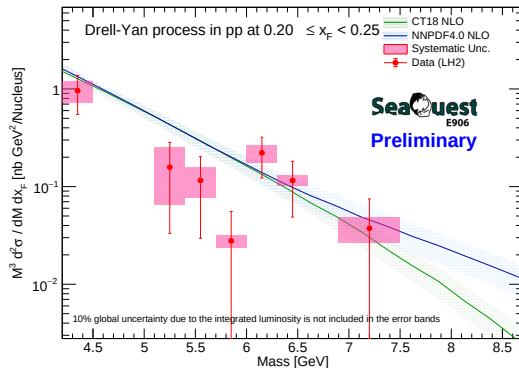


CrossSection LH2 $0.20 \leq x_F < 0.25$ (GeoCenter)

RS67

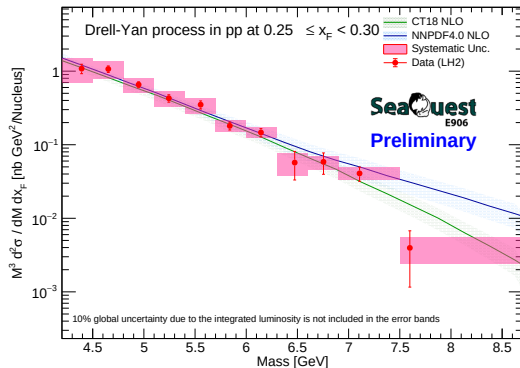


RS59

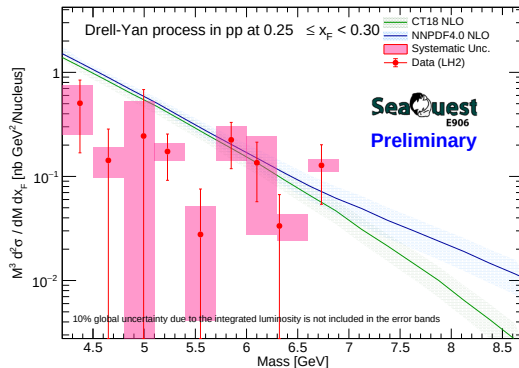


CrossSection LH2 $0.25 \leq x_F < 0.30$ (Centroid)

RS67

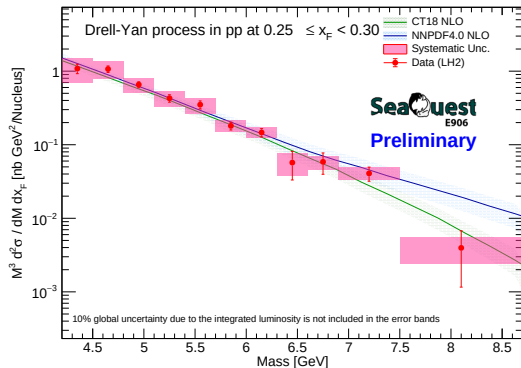


RS59

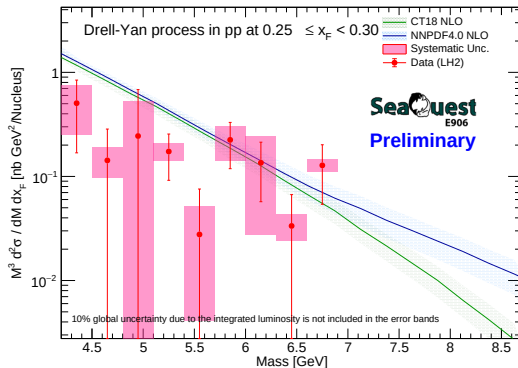


CrossSection LH2 $0.25 \leq x_F < 0.30$ (GeoCenter)

RS67

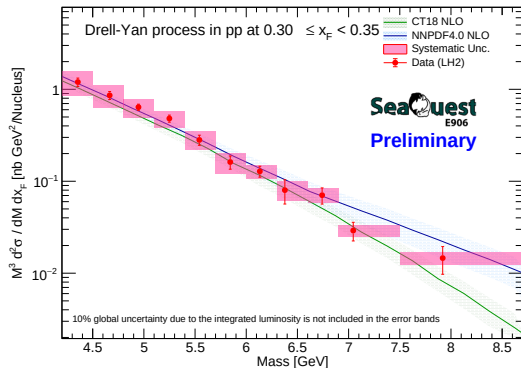


RS59

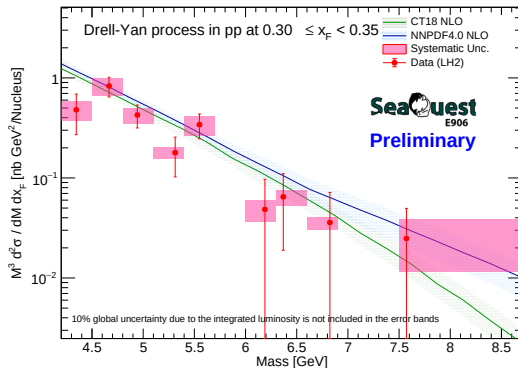


CrossSection LH2 $0.30 \leq x_F < 0.35$ (Centroid)

RS67

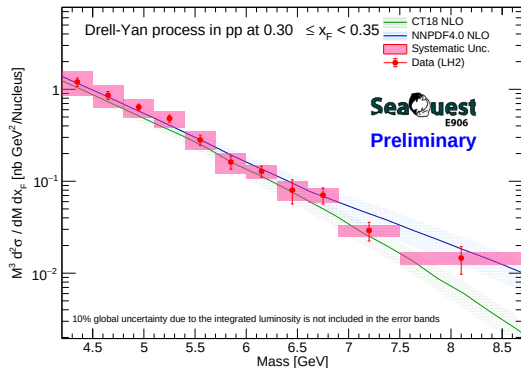


RS59

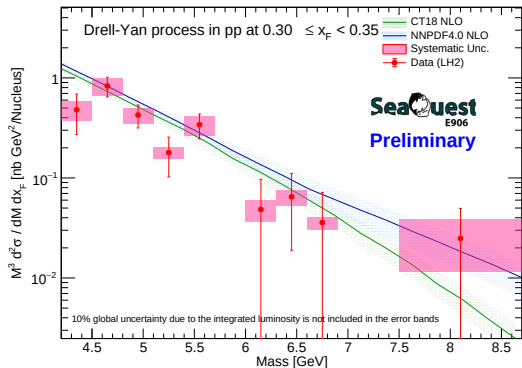


CrossSection LH2 $0.30 \leq x_F < 0.35$ (GeoCenter)

RS67

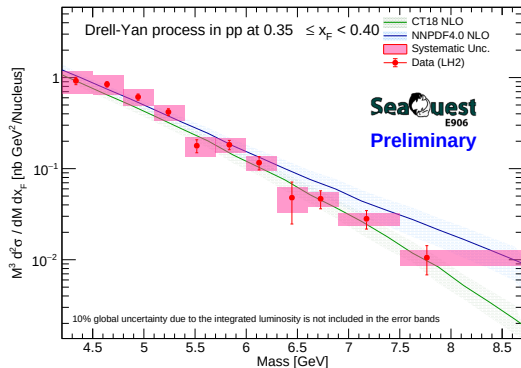


RS59

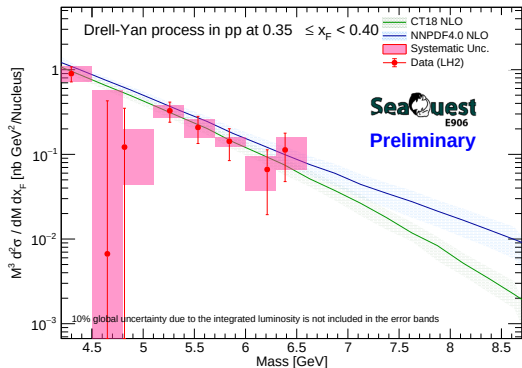


CrossSection LH2 $0.35 \leq x_F < 0.40$ (Centroid)

RS67

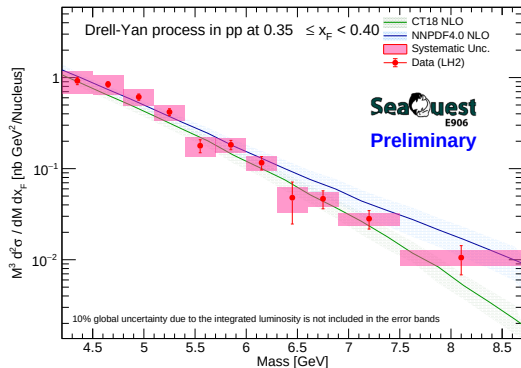


RS59

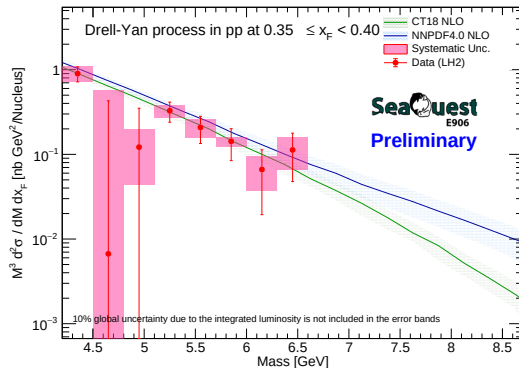


CrossSection LH2 $0.35 \leq x_F < 0.40$ (GeoCenter)

RS67

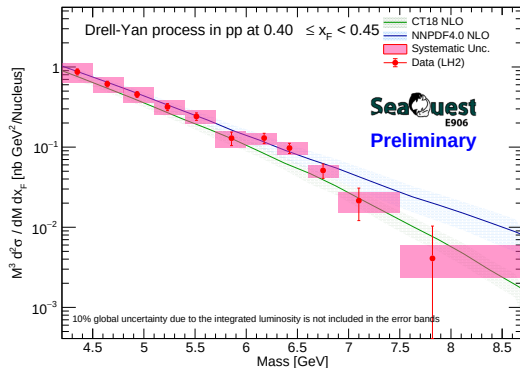


RS59

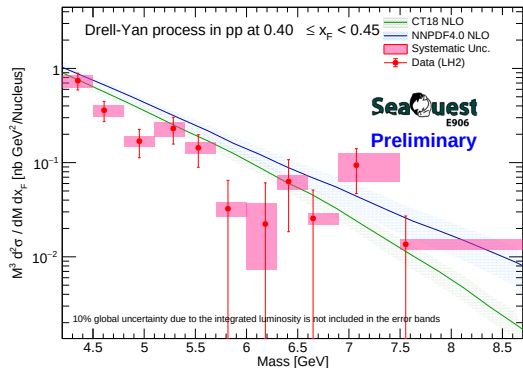


CrossSection LH2 $0.40 \leq x_F < 0.45$ (Centroid)

RS67

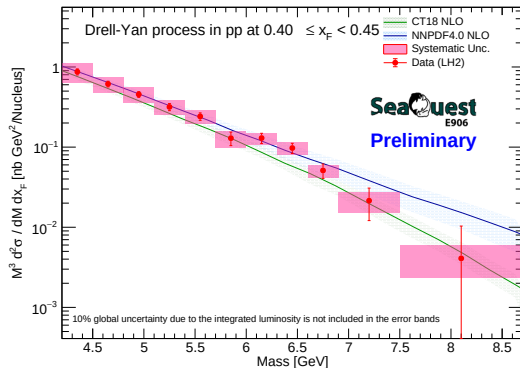


RS59

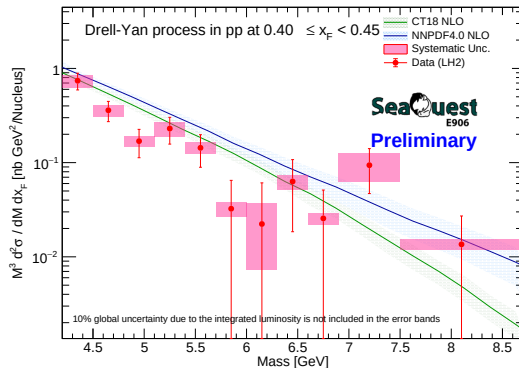


CrossSection LH2 $0.40 \leq x_F < 0.45$ (GeoCenter)

RS67

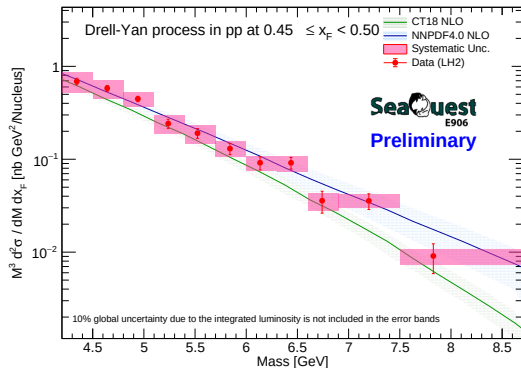


RS59

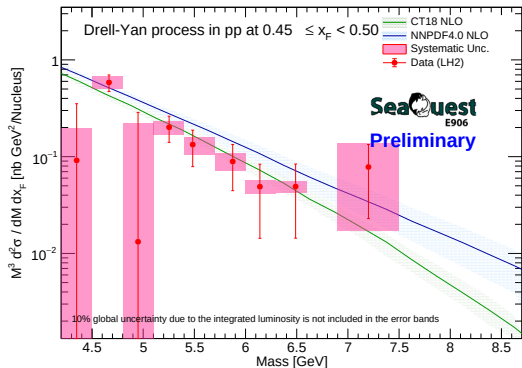


CrossSection LH2 $0.45 \leq x_F < 0.50$ (Centroid)

RS67

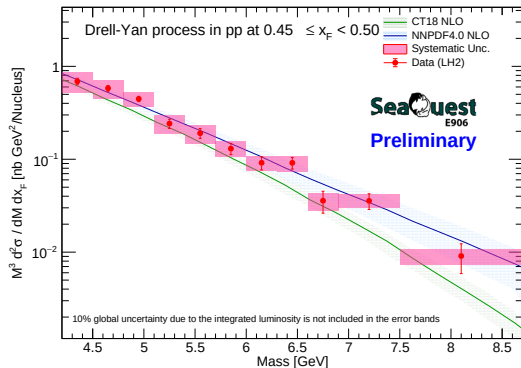


RS59

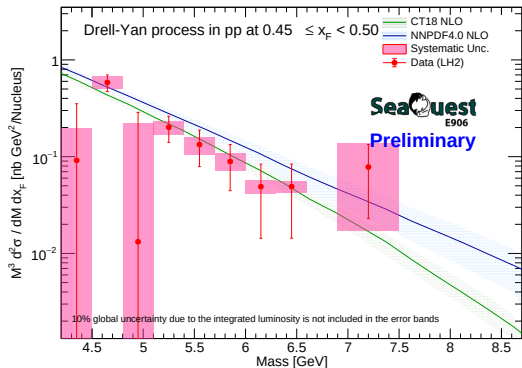


CrossSection LH2 $0.45 \leq x_F < 0.50$ (GeoCenter)

RS67

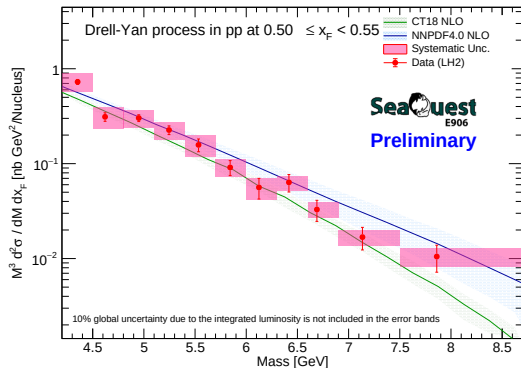


RS59

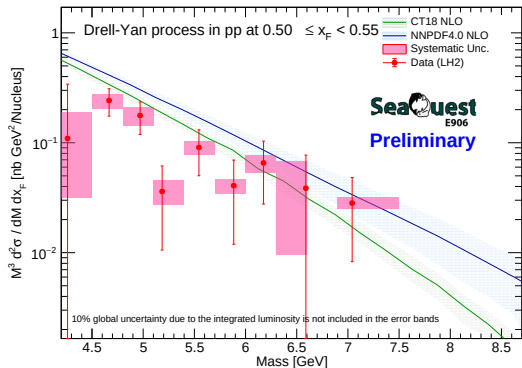


CrossSection LH2 $0.50 \leq x_F < 0.55$ (Centroid)

RS67

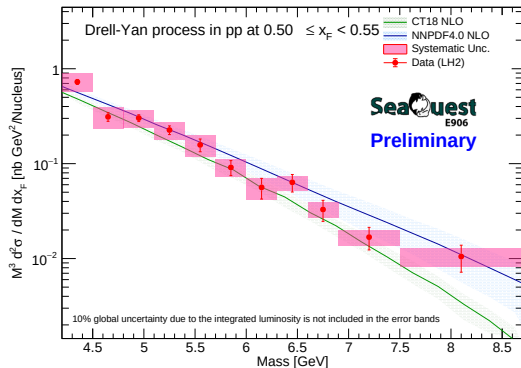


RS59

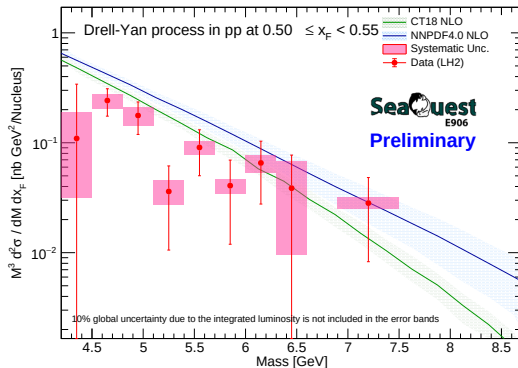


CrossSection LH2 $0.50 \leq x_F < 0.55$ (GeoCenter)

RS67

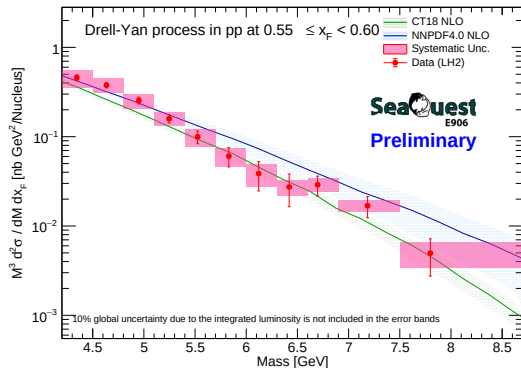


RS59

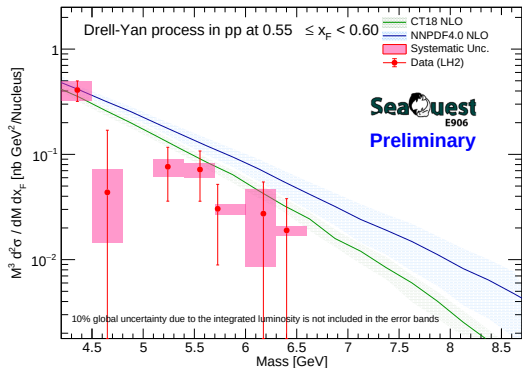


CrossSection LH2 $0.55 \leq x_F < 0.60$ (Centroid)

RS67

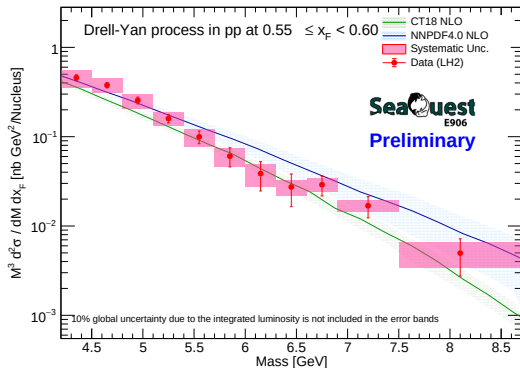


RS59

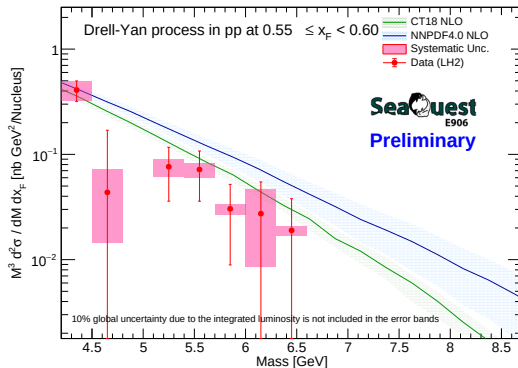


CrossSection LH2 $0.55 \leq x_F < 0.60$ (GeoCenter)

RS67

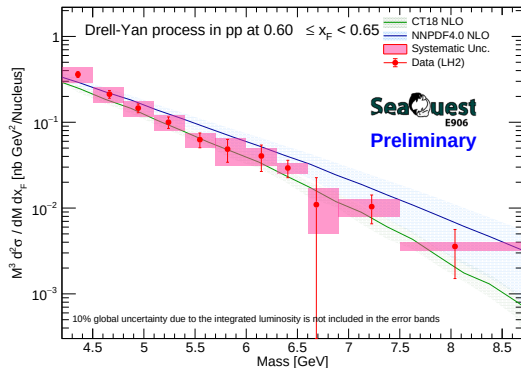


RS59

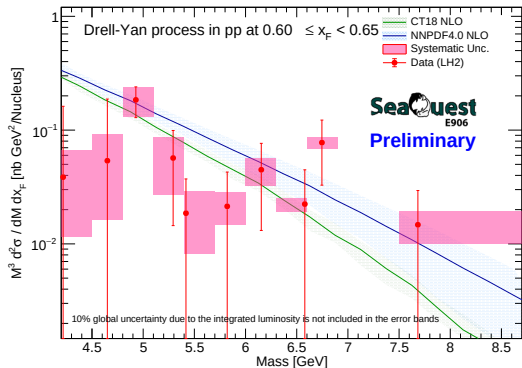


CrossSection LH2 $0.60 \leq x_F < 0.65$ (Centroid)

RS67

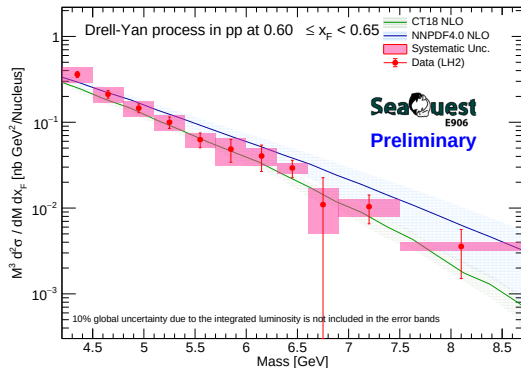


RS59

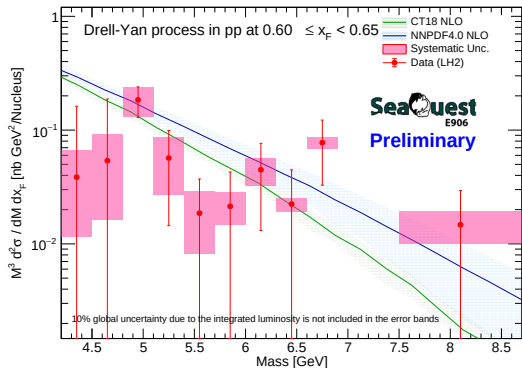


CrossSection LH2 $0.60 \leq x_F < 0.65$ (GeoCenter)

RS67

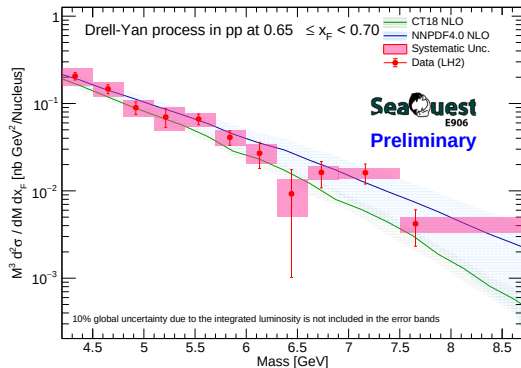


RS59

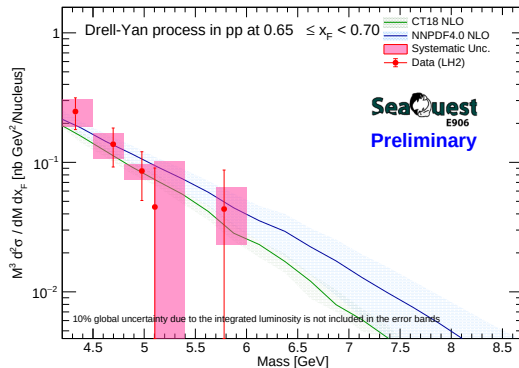


CrossSection LH2 $0.65 \leq x_F < 0.70$ (Centroid)

RS67

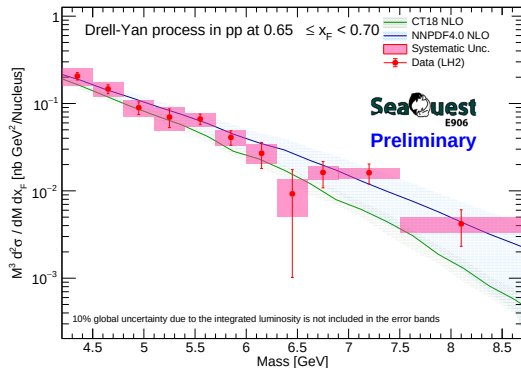


RS59

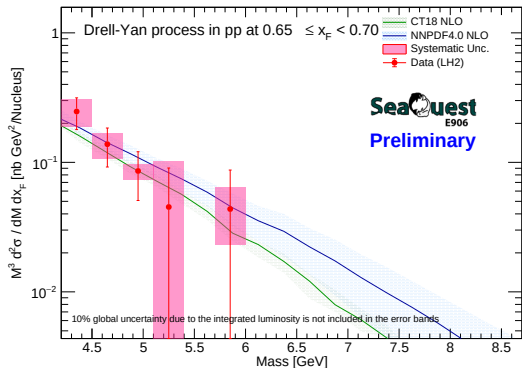


CrossSection LH2 $0.65 \leq x_F < 0.70$ (GeoCenter)

RS67

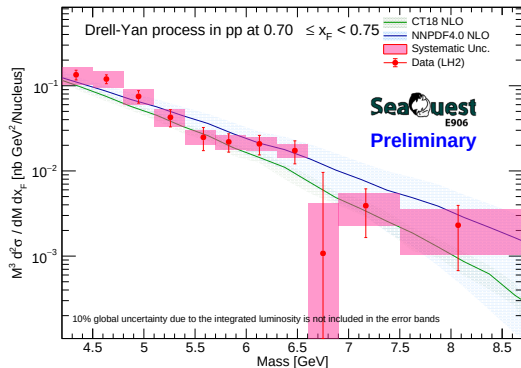


RS59

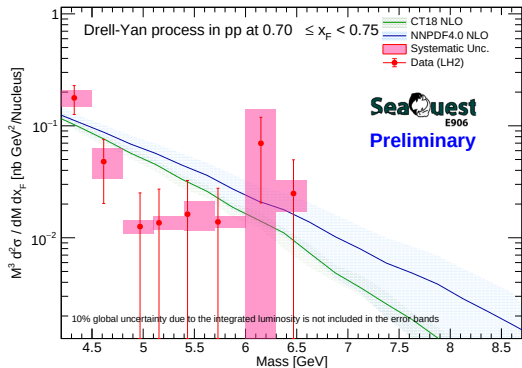


CrossSection LH2 $0.70 \leq x_F < 0.75$ (Centroid)

RS67

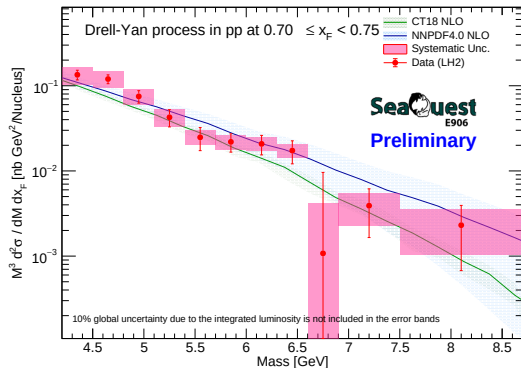


RS59

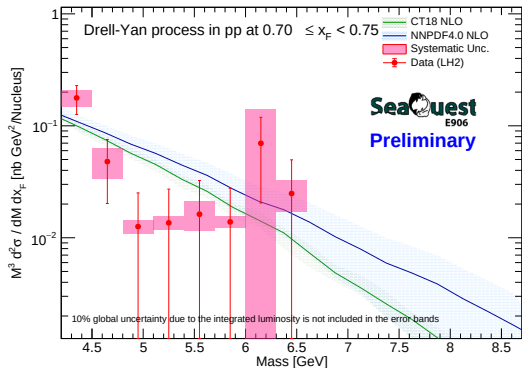


CrossSection LH2 $0.70 \leq x_F < 0.75$ (GeoCenter)

RS67

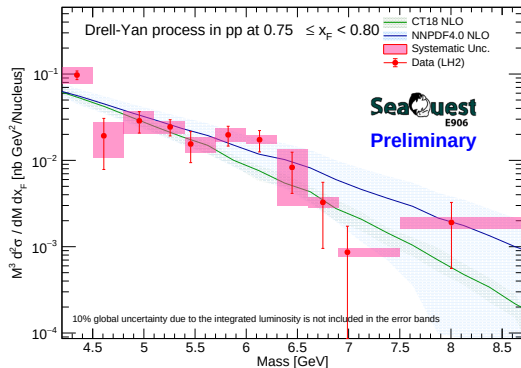


RS59

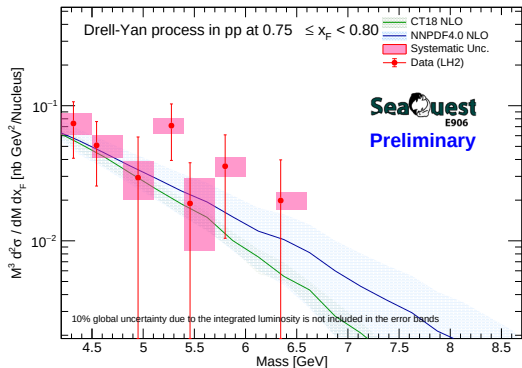


CrossSection LH2 $0.75 \leq x_F < 0.80$ (Centroid)

RS67

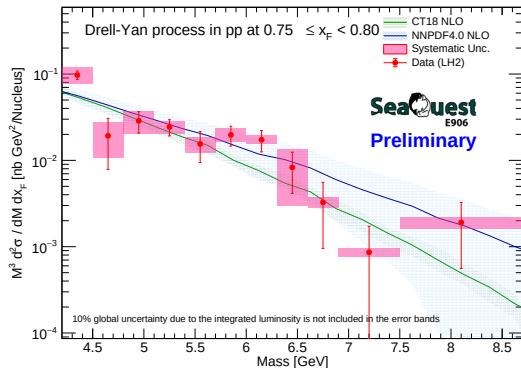


RS59

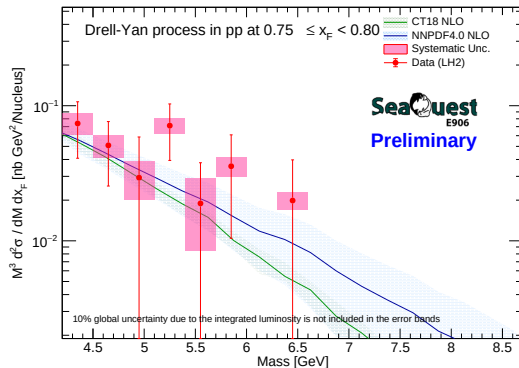


CrossSection LH2 $0.75 \leq x_F < 0.80$ (GeoCenter)

RS67

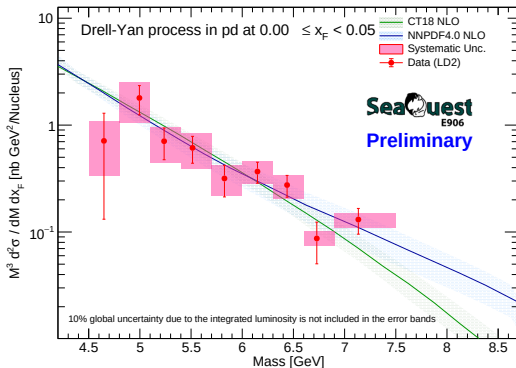


RS59

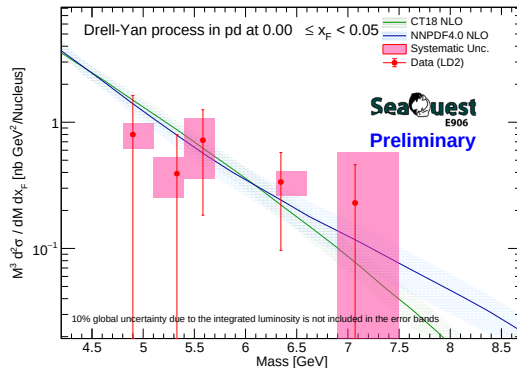


CrossSection LD2 $0.00 \leq x_F < 0.05$ (Centroid)

RS67

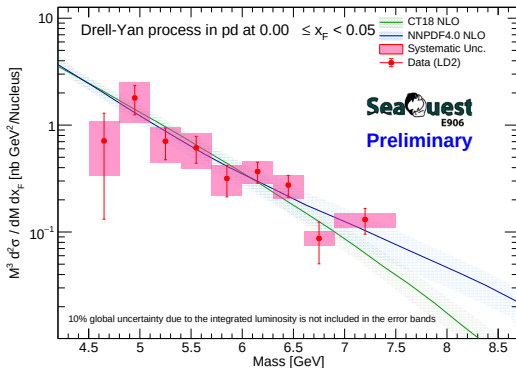


RS59

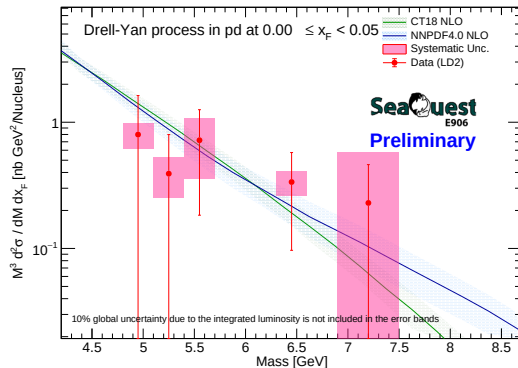


CrossSection LD2 $0.00 \leq x_F < 0.05$ (GeoCenter)

RS67

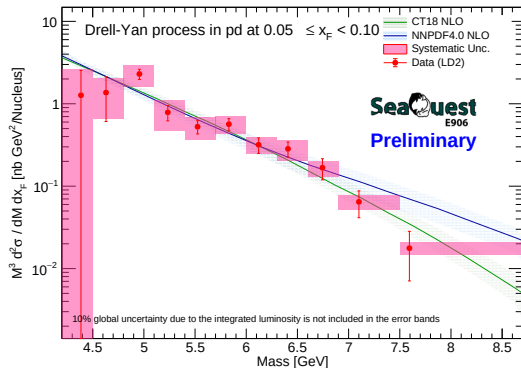


RS59

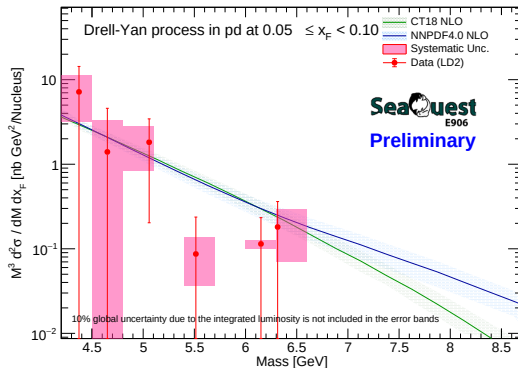


CrossSection LD2 $0.05 \leq x_F < 0.10$ (Centroid)

RS67

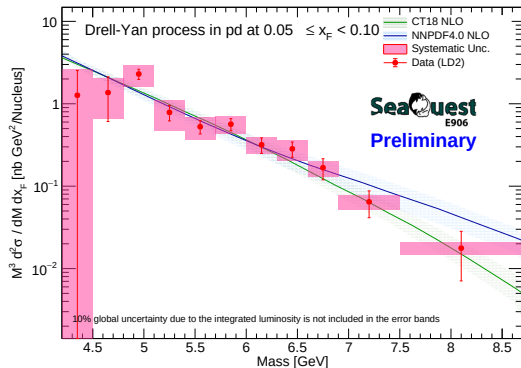


RS59

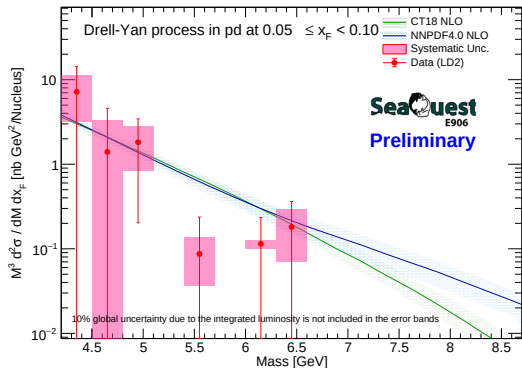


CrossSection LD2 $0.05 \leq x_F < 0.10$ (GeoCenter)

RS67

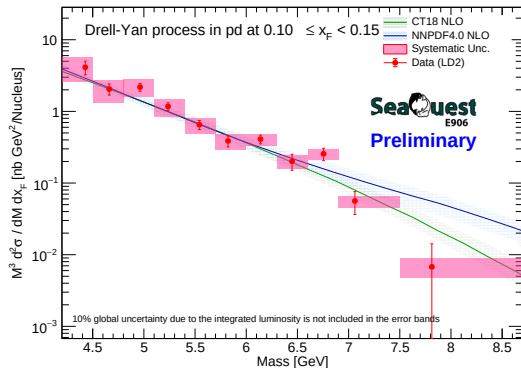


RS59

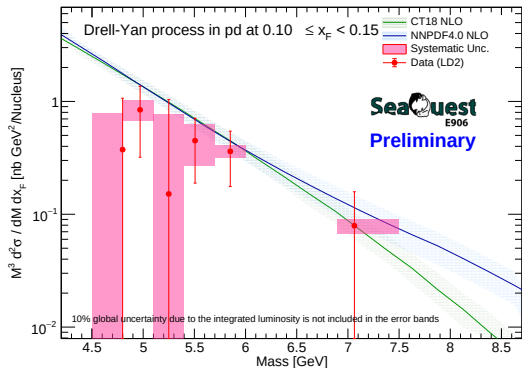


CrossSection LD2 $0.10 \leq x_F < 0.15$ (Centroid)

RS67

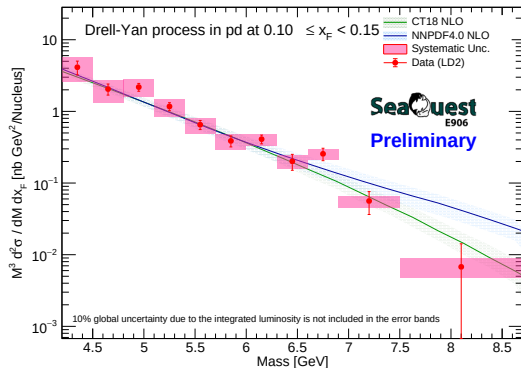


RS59

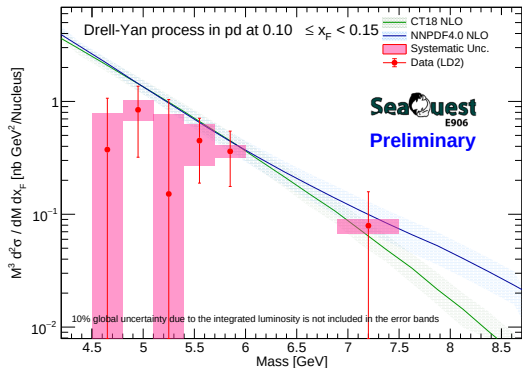


CrossSection LD2 $0.10 \leq x_F < 0.15$ (GeoCenter)

RS67

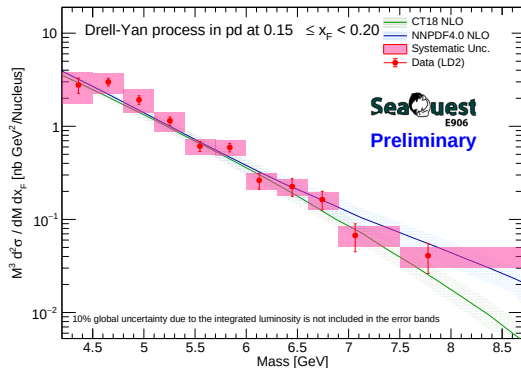


RS59

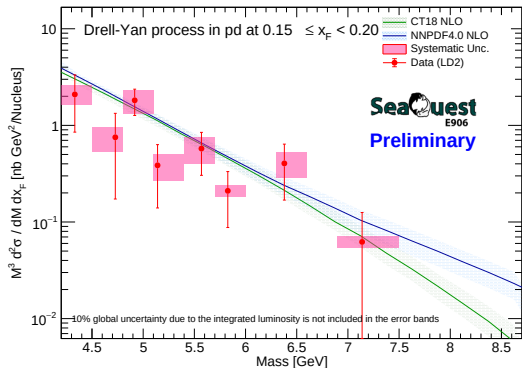


CrossSection LD2 $0.15 \leq x_F < 0.20$ (Centroid)

RS67

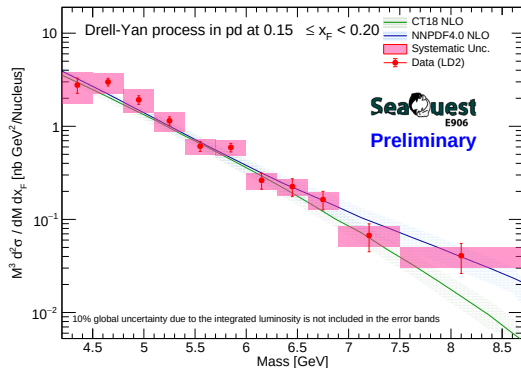


RS59

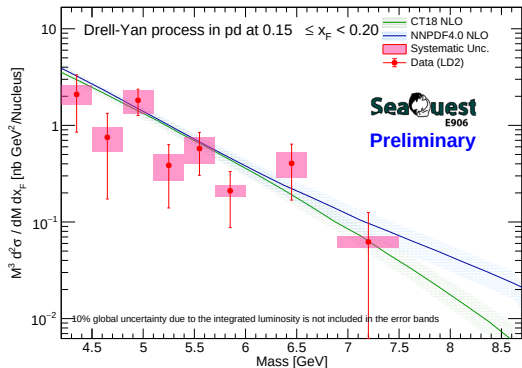


CrossSection LD2 $0.15 \leq x_F < 0.20$ (GeoCenter)

RS67

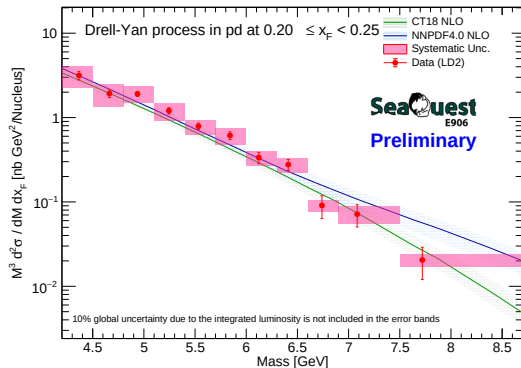


RS59

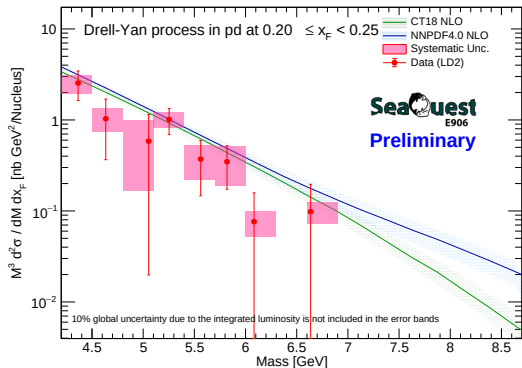


CrossSection LD2 $0.20 \leq x_F < 0.25$ (Centroid)

RS67

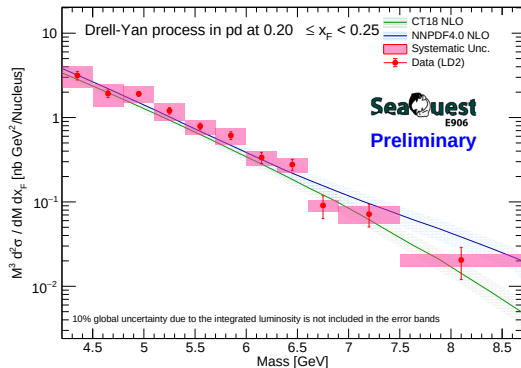


RS59

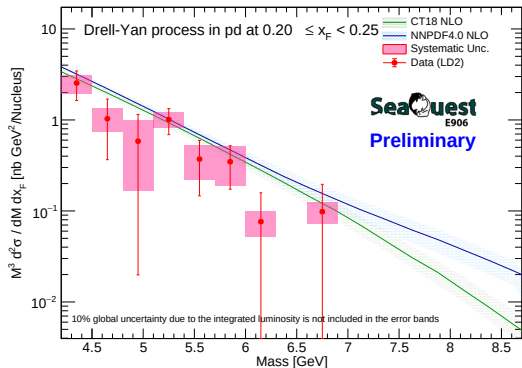


CrossSection LD2 $0.20 \leq x_F < 0.25$ (GeoCenter)

RS67

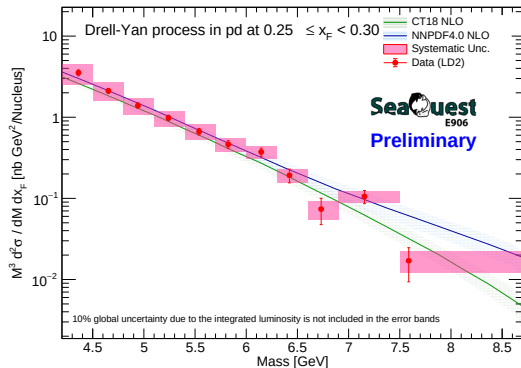


RS59

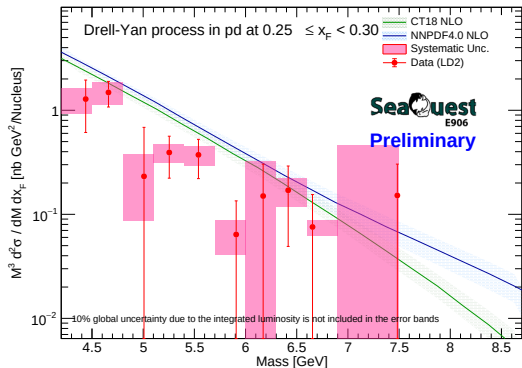


CrossSection LD2 $0.25 \leq x_F < 0.30$ (Centroid)

RS67

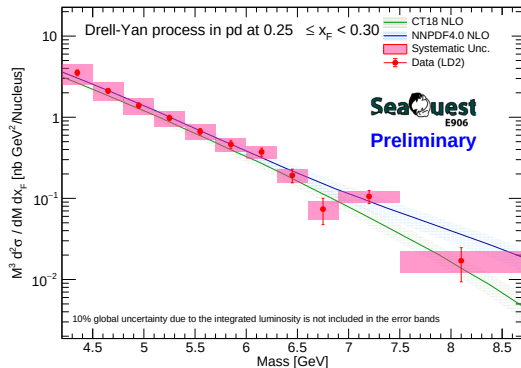


RS59

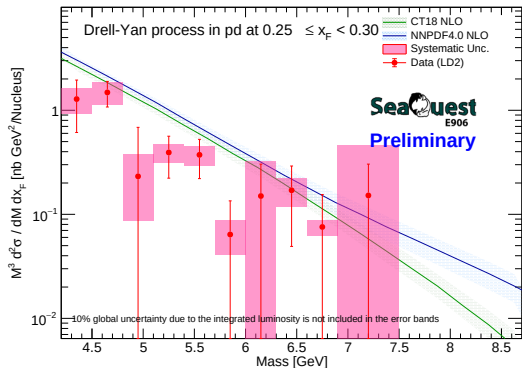


CrossSection LD2 $0.25 \leq x_F < 0.30$ (GeoCenter)

RS67

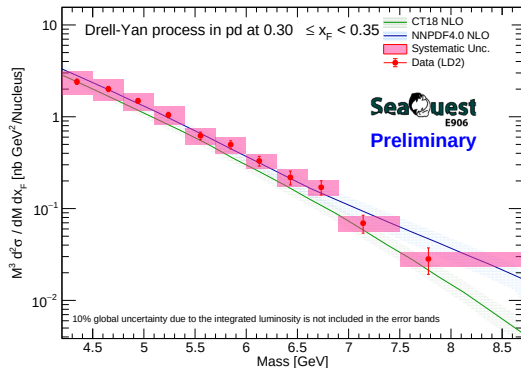


RS59

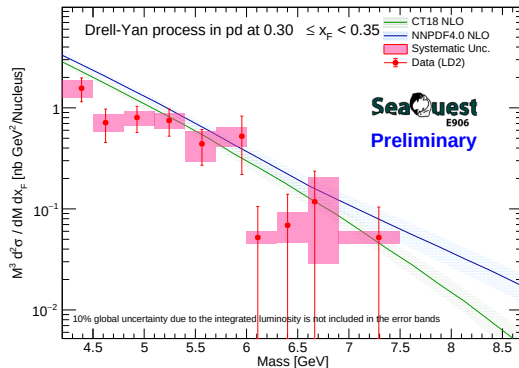


CrossSection LD2 $0.30 \leq x_F < 0.35$ (Centroid)

RS67

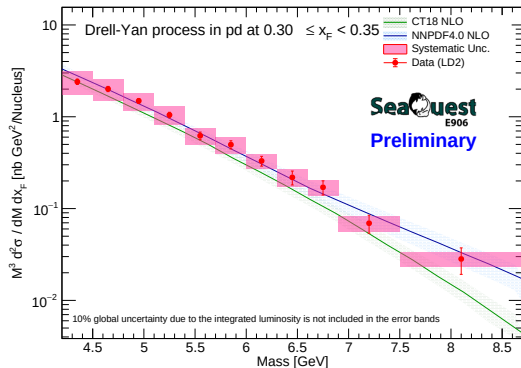


RS59

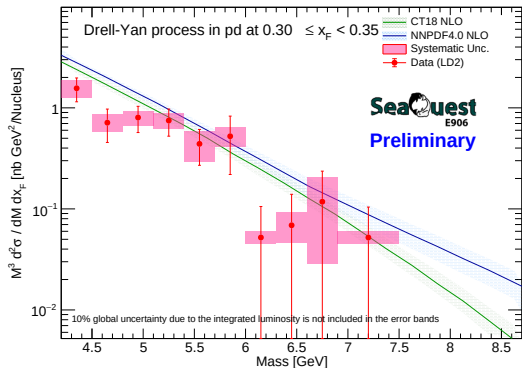


CrossSection LD2 $0.30 \leq x_F < 0.35$ (GeoCenter)

RS67

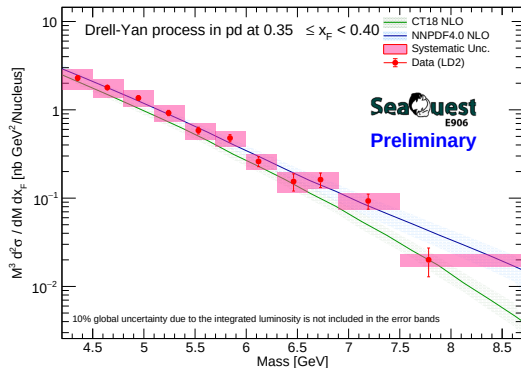


RS59

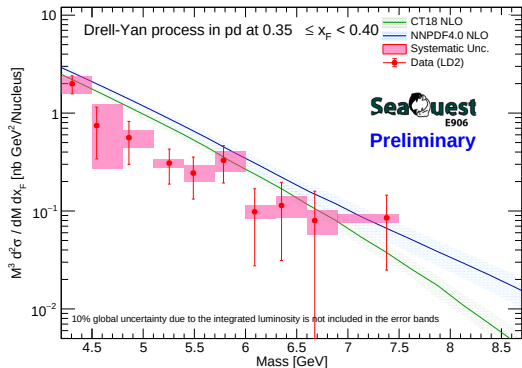


CrossSection LD2 $0.35 \leq x_F < 0.40$ (Centroid)

RS67

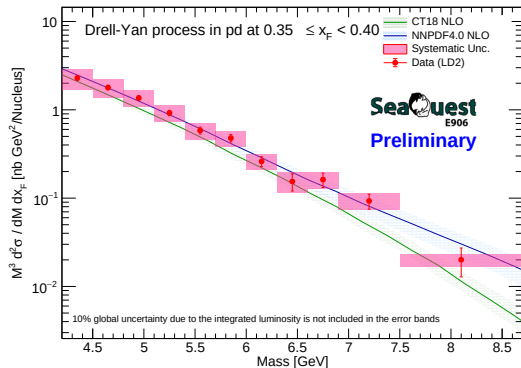


RS59

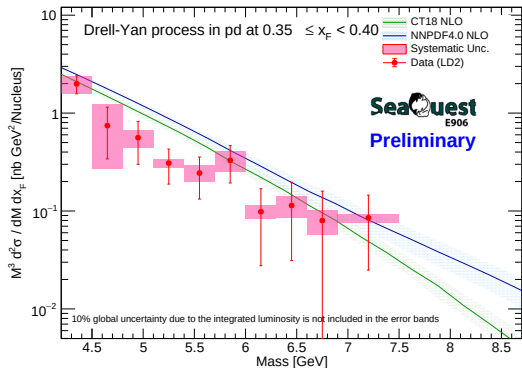


CrossSection LD2 $0.35 \leq x_F < 0.40$ (GeoCenter)

RS67

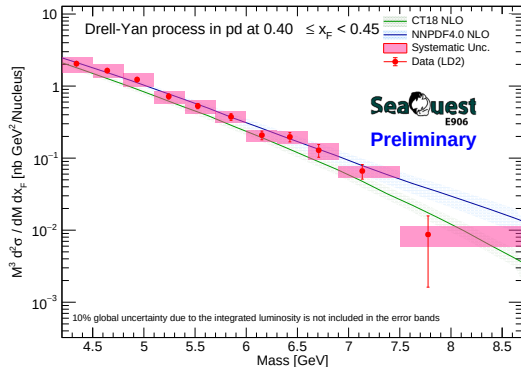


RS59

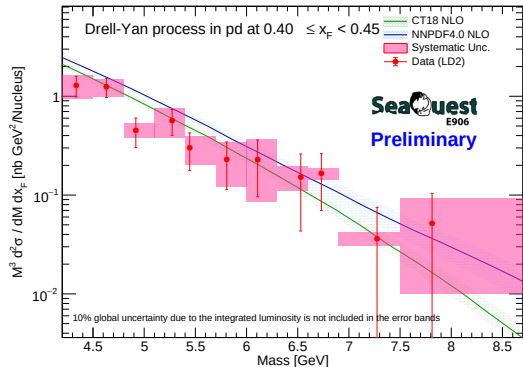


CrossSection LD2 $0.40 \leq x_F < 0.45$ (Centroid)

RS67

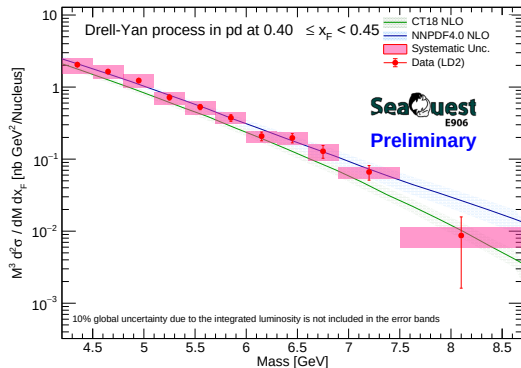


RS59

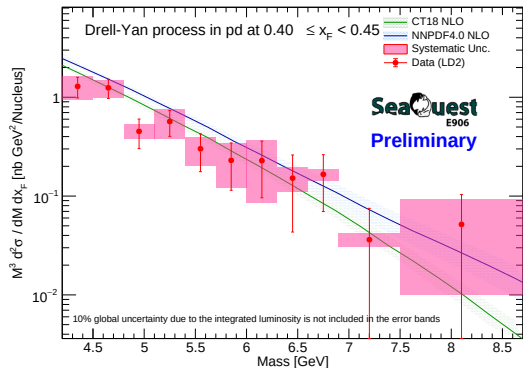


CrossSection LD2 $0.40 \leq x_F < 0.45$ (GeoCenter)

RS67

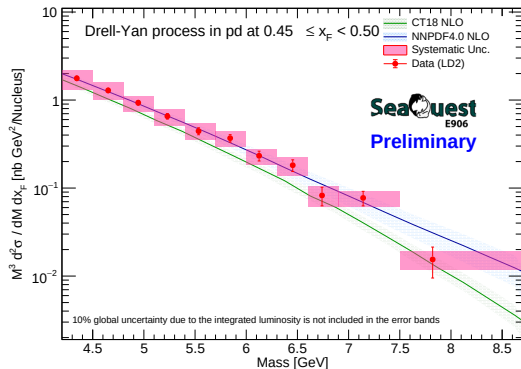


RS59

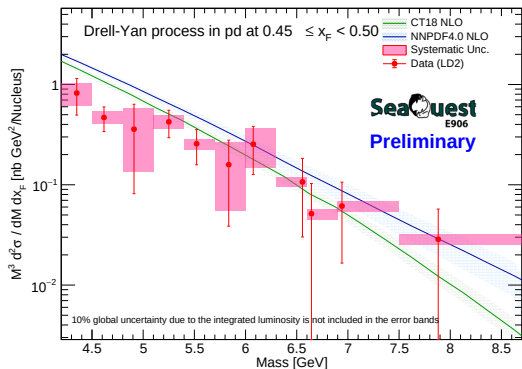


CrossSection LD2 $0.45 \leq x_F < 0.50$ (Centroid)

RS67

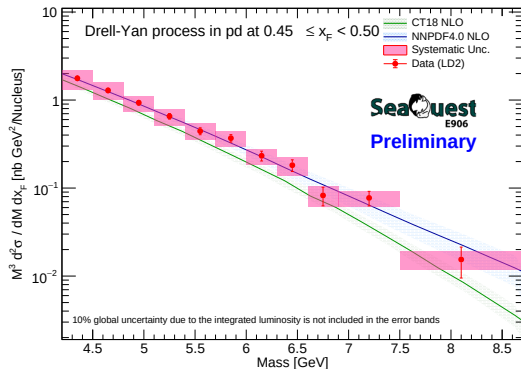


RS59

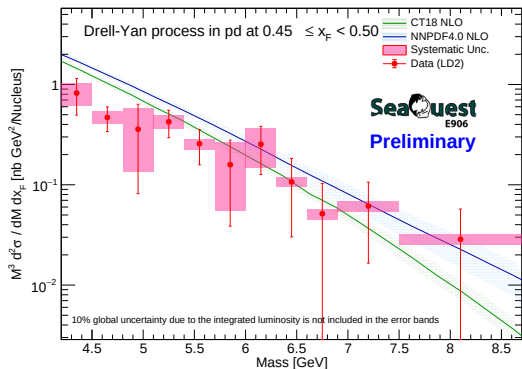


CrossSection LD2 $0.45 \leq x_F < 0.50$ (GeoCenter)

RS67

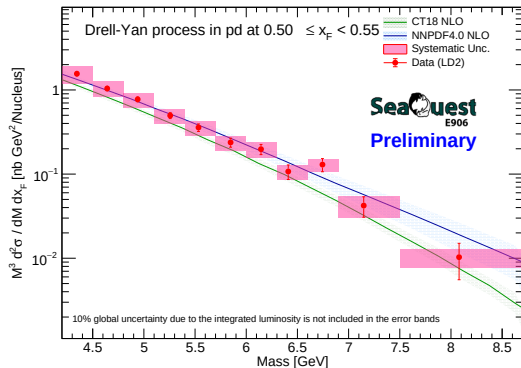


RS59

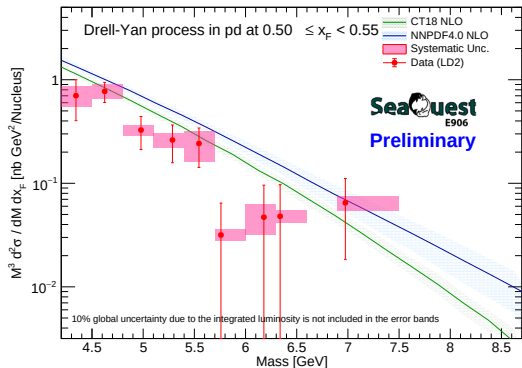


CrossSection LD2 $0.50 \leq x_F < 0.55$ (Centroid)

RS67

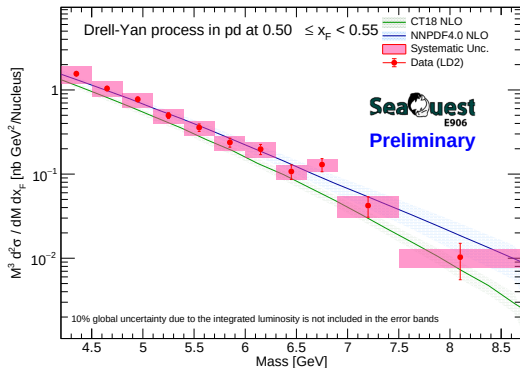


RS59

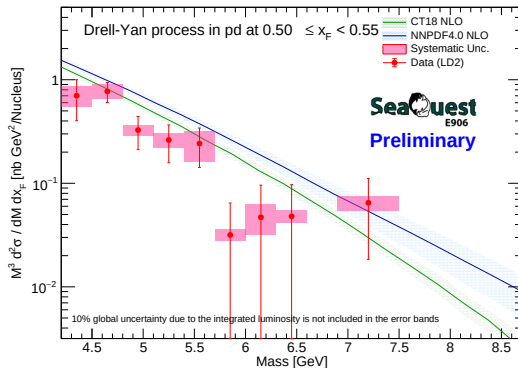


CrossSection LD2 $0.50 \leq x_F < 0.55$ (GeoCenter)

RS67

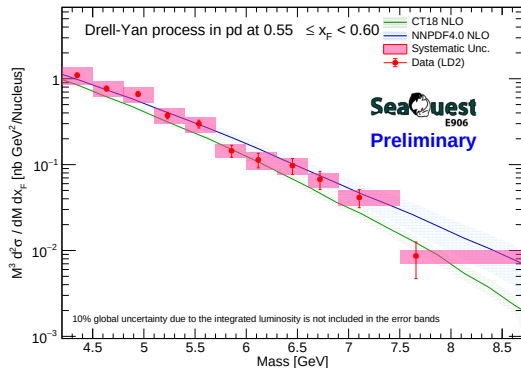


RS59

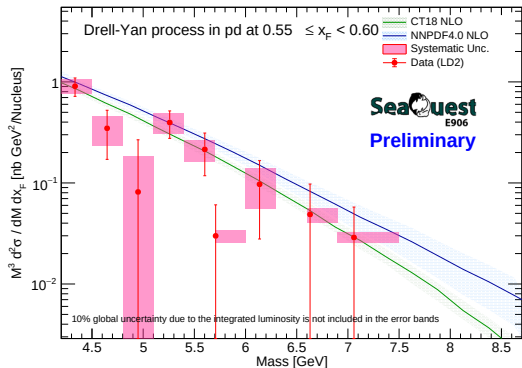


CrossSection LD2 $0.55 \leq x_F < 0.60$ (Centroid)

RS67

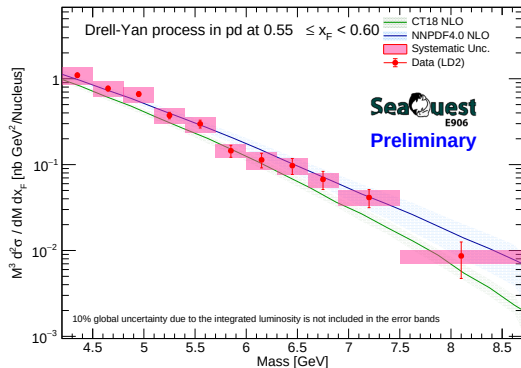


RS59

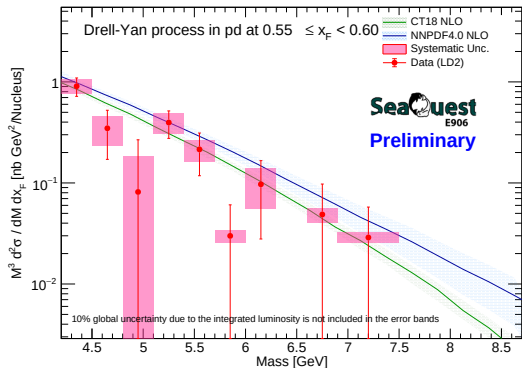


CrossSection LD2 $0.55 \leq x_F < 0.60$ (GeoCenter)

RS67

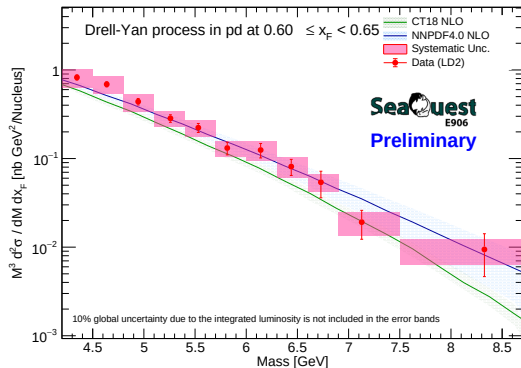


RS59

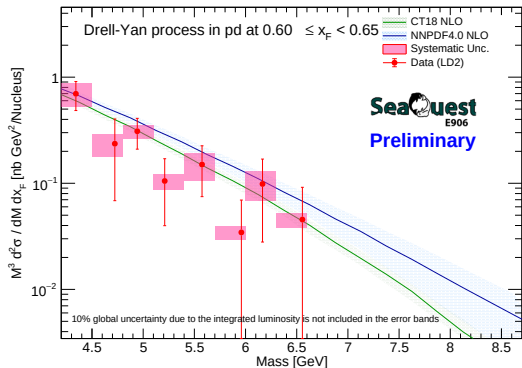


CrossSection LD2 $0.60 \leq x_F < 0.65$ (Centroid)

RS67

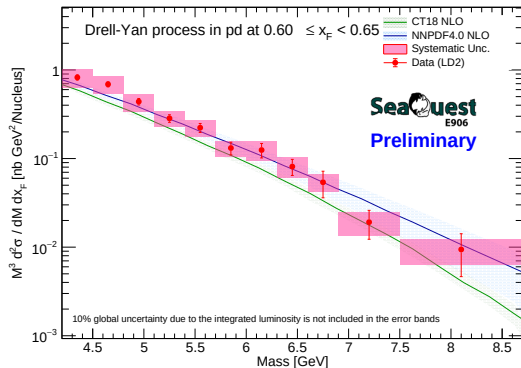


RS59

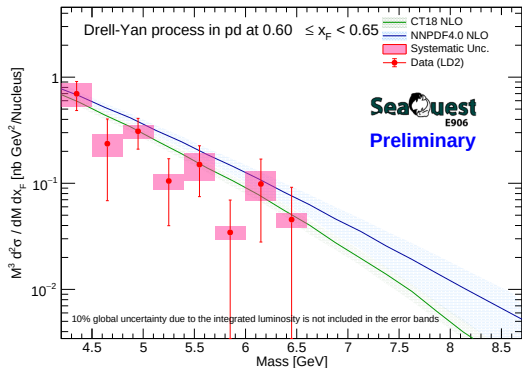


CrossSection LD2 $0.60 \leq x_F < 0.65$ (GeoCenter)

RS67

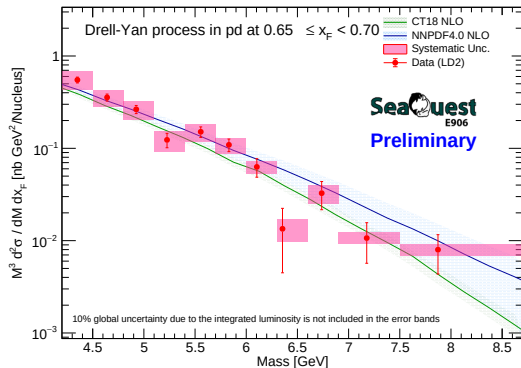


RS59

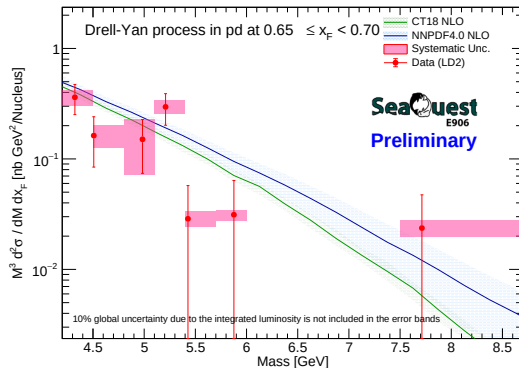


CrossSection LD2 $0.65 \leq x_F < 0.70$ (Centroid)

RS67

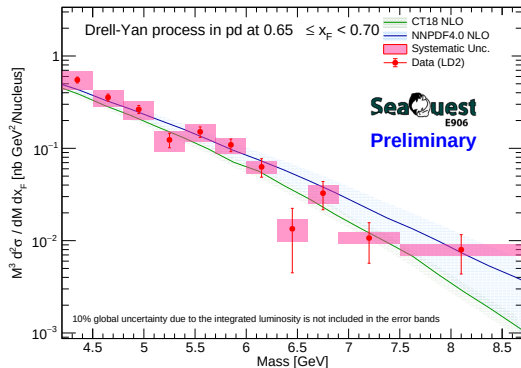


RS59

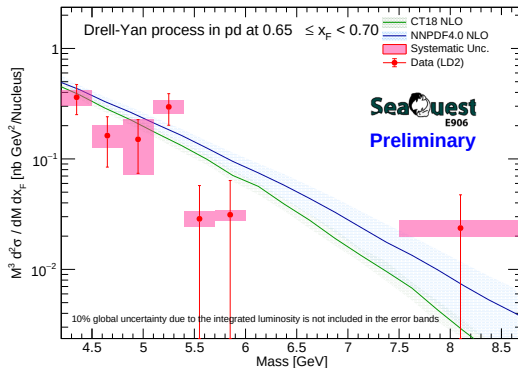


CrossSection LD2 $0.65 \leq x_F < 0.70$ (GeoCenter)

RS67

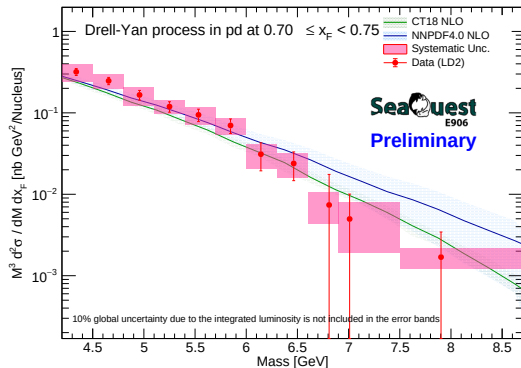


RS59

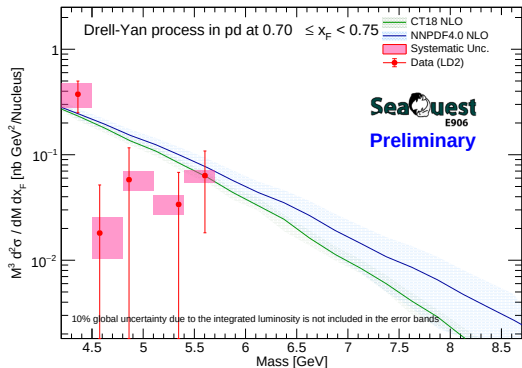


CrossSection LD2 $0.70 \leq x_F < 0.75$ (Centroid)

RS67

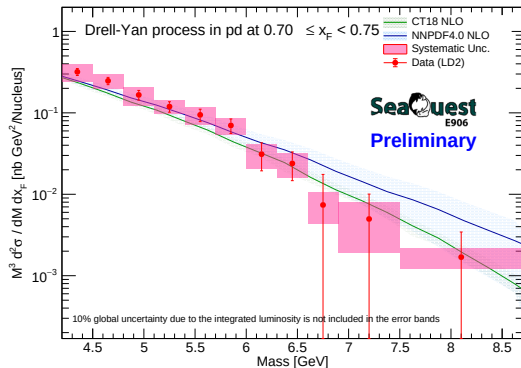


RS59

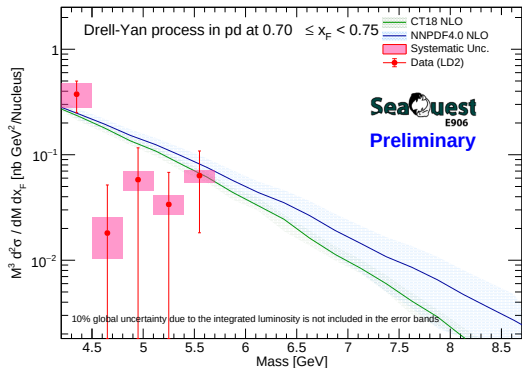


CrossSection LD2 $0.70 \leq x_F < 0.75$ (GeoCenter)

RS67

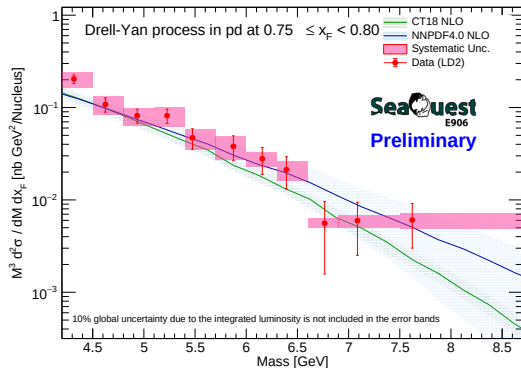


RS59

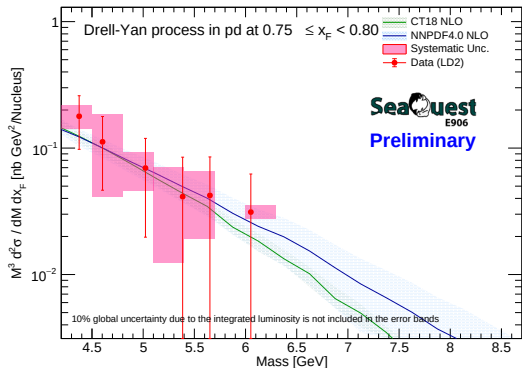


CrossSection LD2 $0.75 \leq x_F < 0.80$ (Centroid)

RS67

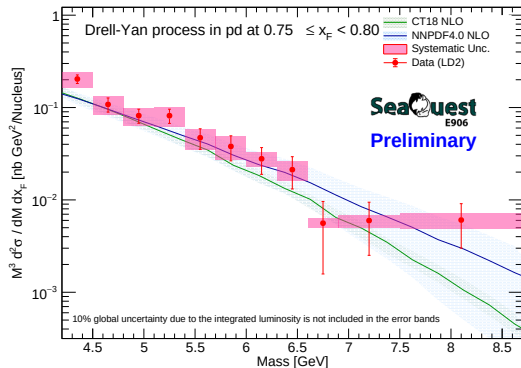


RS59



CrossSection LD2 $0.75 \leq x_F < 0.80$ (GeoCenter)

RS67



RS59

